MATA31 - Calculus I for Mathematical Sciences

Review Slides

Outline of Topics.

- Sets and notation ($\in$, $\notin$, $\subseteq$, $\emptyset$, $\cup$, $\cap$)
- Numbers ($\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$) and intervals
- Properties of $\mathbb{R}$
- Properties of inequalities ($<$, $\leq$, $>$, $\geq$)
- Properties of absolute values ($|\bullet|$)
- Lines and conics (parabola, circle, ellipse, hyperbola)
- Trigonometry ($\sin x$, $\cos x$, $\tan x$)
- Functions (domain, range, VLT, common types, even/odd, inc/dec, composition)
- Exponential functions and logarithms (e^x, $\ln x$)
- Inverses and inverse trig ($\arcsin x$, $\arccos x$, $\arctan x$)

Sets

Definition

A $\boxed{\textbf{set}}$ is a collection of elements where

- the elements are distinct (distinguishable),
- the order of the elements does **not** matter,

(* and other technicalities found in a logic course, e.g., a set cannot be a member of itself.)

Example

Curly braces denote sets:

- $S = \{1, 2, 3, 4\}$ is a set of four elements, namely the numbers 1, 2, 3 and 4.
- $A = \{\text{red}, \text{green}, \text{blue}\}$ is a set of three elements.
- $B = \{\{1, 2\}, \{3, 4\}, \{5, 6\}\}$ is a set of three elements (the elements are also sets).

Definition

The $\boxed{\textbf{empty set}}$ is the set containing no elements. We denote it by $\emptyset$ or $\{\}$.

Set Notation

Notation

- If x is an element of a set S, then we write $\boxed{x \in S}$.
- If x is **not** an element of a set S, then we write $\boxed{x \notin S}$.

Example

Let $A = \{1, 2, 3, 4\}$, $B = \{1, 2, 4\}$ and $C = \{4, 5\}$.
Then $2 \in A$, $2 \in B$ and $2 \notin C$.

Notation

Sets are often represented using set-builder notation:

$$\boxed{S = \left\{x \in \text{category} : \text{constraints on } x\right\}} \quad \text{or} \quad \boxed{S = \left\{x \in \text{category} \mid \text{constraints on } x\right\}}$$

The symbols $\boxed{:}$ and $\boxed{\mid}$ mean "**such that**". (Only use these in set-builder notation.)

Subsets

Definition

Let A and B be sets. We say A is a $\boxed{\textbf{subset}}$ of B if every element of A is an element of B.

We denote this by $A \subseteq B$ (or by $B \supseteq A$).

Example

Let $A = \{1, 2, 3, 4\}$, $B = \{1, 2, 4\}$ and $C = \{4, 5\}$.

Then $B \subseteq A$ and $C \not\subseteq A$ (since $5 \in C$ but $5 \notin A$).

Example

Let $S = \{1, 2, 3, 4\}$. How many subsets does S have?

Solution.
The answer is 16. We list all of the subsets:

- $\{\}$,
- $\{1\}$, $\{2\}$, $\{3\}$, $\{4\}$,
- $\{1, 2\}$, $\{1, 3\}$, $\{1, 4\}$, $\{2, 3\}$, $\{2, 4\}$, $\{3, 4\}$,
- $\{1, 2, 3\}$, $\{1, 2, 4\}$, $\{1, 3, 4\}$, $\{2, 3, 4\}$,
- $\{1, 2, 3, 4\}$.

Union and Intersection

Definition: Union and Intersection

Let A and B be sets.

- The **union** of A and B is the set $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$.
- The **intersection** of A and B is the set $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$.

Mnemonic: The union symbol $\cup$ looks like the letter U standing for **u**nion.

Example

Let $A = \{1, 2, 3, 4\}$, $B = \{1, 2, 4\}$ and $C = \{4, 5\}$.

Then $A \cup B = \{1, 2, 3, 4\}$ and $A \cap C = \{4\}$.

Numbers

Symbol	Description	Set
$\mathbb{N}$	The **natural numbers**	$\{1, 2, 3, \ldots\}$ (some textbooks include 0)
$\mathbb{Z}$	The **integers**	$\{\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots\}$
$\mathbb{Q}$	The **rational numbers**	$\left\{\frac{p}{q} : p, q \in \mathbb{Z}, q \neq 0\right\}$
$\mathbb{R}$	The **real numbers**	Can be written using a finite or infinite **decimal expansion**
$\mathbb{C}$	The **complex numbers**	Allows us to solve equations like $x^2 + 1 = 0$

The five sets of numbers above satisfy:

$$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C},$$

that is, all natural numbers are integers, all integers are rational numbers, etc.

Example

- The numbers $-\frac{3}{4}$, 2.647, 17, $0.\overline{7}$ are all rational numbers.
- Note that 2.647 can be written as a fraction:

$$2.647 = 2.647 \times \frac{1000}{1000} = \frac{2647}{1000}.$$

- Rationals have a decimal expansion that is either terminating or repeating.
- Real numbers not ending in a terminating or repeating pattern are called **irrational**.

Example

Well-known irrational numbers include:

- $\sqrt{2}$ $\quad(\approx 1.41421\ldots)$
- π $\quad(\approx 3.14159\ldots)$
- e $\quad(\approx 2.71828\ldots)$

Question

Can you **prove** that $\sqrt{2}$ is irrational?

Open Intervals

Definition

Let $a < b$ where $a, b \in \mathbb{R}$. Then $\boxed{(a, b)}$ is the open interval from a to b and consists of all the real numbers between a and b but **not** including a or b:

$$(a, b) = \{x \in \mathbb{R} \mid a < x < b\}.$$

- On the number line we depict this as:

- Circles at a and b are **open circles**, indicating a and b are omitted.
- **Mnemonic:** Round brackets () and open circles form an **O** shape for "**O**pen and **O**mit".

Closed Intervals

Definition

Let $a < b$ where $a, b \in \mathbb{R}$. Then $\boxed{[a, b]}$ is the closed interval from a to b and consists of all the real numbers between a and b **including** a and b:

$$[a, b] = \{x \in \mathbb{R} \mid a \leq x \leq b\}.$$

- On the number line we depict this as:

- Circles at a and b are **closed circles**, indicating a and b are included.

Types of Intervals

Let $a < b$ where $a, b \in \mathbb{R}$. Other types of intervals include:

$(a, b) = \{x \in \mathbb{R} \,|\, a < x < b\}$

$[a, b] = \{x \in \mathbb{R} \,|\, a \leq x \leq b\}$

$[a, b) = \{x \in \mathbb{R} \,|\, a \leq x < b\}$

$(a, b] = \{x \in \mathbb{R} \,|\, a < x \leq b\}$

$(a, \infty) = \{x \in \mathbb{R} \,|\, x > a\}$

$[a, \infty) = \{x \in \mathbb{R} \,|\, x \geq a\}$

$(-\infty, b) = \{x \in \mathbb{R} \,|\, x < b\}$

$(-\infty, b] = \{x \in \mathbb{R} \,|\, x \leq b\}$

Properties of the Real Numbers

The following theorems about real numbers can be proven using axioms not covered in this course.

Theorem

Let $a, b \in \mathbb{R}$.
Then $ab = 0$ if and only if $a = 0$ <u>or</u> $b = 0$.
Then $ab \neq 0$ if and only if $a \neq 0$ <u>and</u> $b \neq 0$.

An expression is called a **real expression** if when calculated it results in a real number.

Theorem

Let A and B be real expressions. Then $AB = 0$ if and only if $A = 0$ or $B = 0$.

Theorem

Let $a, b, c, d \in \mathbb{R}$ with $b, c, d \neq 0$. Then

- $\dfrac{a}{b} + \dfrac{c}{d} = \dfrac{ad + bc}{bd}$
- $\dfrac{a}{b} - \dfrac{c}{d} = \dfrac{ad - bc}{bd}$
- $\left(\dfrac{a}{b}\right)\left(\dfrac{c}{d}\right) = \dfrac{ac}{bd}$
- $\dfrac{(a/b)}{(c/d)} = \left(\dfrac{a}{b}\right)\left(\dfrac{d}{c}\right) = \dfrac{ad}{bc}$

Solving Quadratics

Theorem

The Quadratic Formula

The solutions to $ax^2 + bx + c = 0$ (with $a \neq 0$) are: $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

Can **prove** the quadratic formula? (Try completing the square.)

Example

Solve $x^2 - 9 = 0$.

Solution:

- **Method 1:** Factor as a difference of squares:

$$(x-3)(x+3) = 0$$

  Thus, $x = 3$ or $x = -3$ (by theorem on previous slide).
- **Method 2:** Use the Quadratic Formula with $a = 1$, $b = 0$ and $c = -9$ to get:

$$x = \frac{0 \pm \sqrt{-4(1)(-9)}}{2} = \pm 3.$$

- **Method 3:** Solving for x gives:

$$x^2 - 9 = 0 \quad \Longrightarrow \quad x^2 = 9 \quad \Longrightarrow \quad x = \pm\sqrt{9} \quad \Longrightarrow \quad x = \pm 3.$$

We review **long division** of polynomials on this slide:

Example: Compute $\dfrac{x^3 - 2x^2 - 4}{x - 3}$, that is, divide $x - 3$ into $x^3 - 2x^2 - 4$.

- First write the dividend as $x^3 - 2x^2 + 0x - 4$. (that is, include 0's).
- Divide the first term of the dividend by the highest term of the divisor:

$$\begin{array}{r|l} & x^2 \\ \hline x-3\ ) & x^3 - 2x^2 + 0x - 4 \end{array} \qquad \text{Here } x^3 \div x = x^2$$

- Multiply the divisor by this result and write it below:

$$\begin{array}{r|l} & x^2 \\ \hline x-3\ ) & x^3 - 2x^2 + 0x - 4 \\ & x^3 - 3x^2 \end{array} \qquad \text{Here } x^2 \cdot (x-3) = x^3 - 3x^2$$

- Subtract and write the result below:

$$\begin{array}{r|l} & x^2 \\ \hline x-3\ ) & x^3 - 2x^2 + 0x - 4 \\ & x^3 - 3x^2 \\ \hline & \qquad +x^2 + 0x - 4 \end{array}$$

- (Repeat) Divide, multiply then subtract:

$$\begin{array}{r|l} & x^2 \ + \ x \\ \hline x-3\ ) & x^3 - 2x^2 + 0x - 4 \\ & x^3 - 3x^2 \\ \hline & \qquad +x^2 + 0x - 4 \end{array} \rightarrow \begin{array}{r|l} & x^2 \ + \ \mathbf{x} \\ \hline x-3\ ) & x^3 - 2x^2 + 0x - 4 \\ & x^3 - 3x^2 \\ \hline & \qquad +x^2 + 0x - 4 \\ & \qquad +x^2 - 3x \end{array} \rightarrow \begin{array}{r|l} & x^2 \ + \ x \\ \hline x-3\ ) & x^3 - 2x^2 + 0x - 4 \\ & x^3 - 3x^2 \\ \hline & \qquad +x^2 + 0x - 4 \\ & \qquad +x^2 - 3x \\ \hline & \qquad\qquad 3x - 4 \end{array}$$

CONTINUED:

- (Repeat) We divide:

$$\begin{array}{r l}
 & x^2 \quad + \quad x \quad + \quad \mathbf{3} \\ \cline{2-2}
x-3 &)x^3 - 2x^2 + 0x - 4 \\
 & \underline{x^3 - 3x^2} \\
 & +x^2 + 0x - 4 \\
 & \underline{+x^2 - 3x} \\
 & 3x - 4
\end{array}$$

- Then multiply and subtract:

$$\begin{array}{r l}
 & x^2 \quad + \quad x \quad + \quad \mathbf{3} \\ \cline{2-2}
x-3 &)x^3 - 2x^2 + 0x - 4 \\
 & \underline{x^3 - 3x^2} \\
 & +x^2 + 0x - 4 \\
 & \underline{+x^2 - 3x} \\
 & 3x - 4 \\
 & \mathbf{3x\text{-}9}
\end{array}
\quad \rightarrow \quad
\begin{array}{r l}
 & x^2 \quad + \quad x \quad + \quad \mathbf{3} \\ \cline{2-2}
x-3 &)x^3 - 2x^2 + 0x - 4 \\
 & \underline{x^3 - 3x^2} \\
 & +x^2 + 0x - 4 \\
 & \underline{+x^2 - 3x} \\
 & 3x - 4 \\
 & \underline{3x - 9} \\
 & \mathbf{5}
\end{array}$$

- The **quotient** is $x^2 + x + 3$ and the **remainder** is 5.
- Therefore:

$$\frac{x^3 - 2x^2 - 4}{x - 3} = (x^2 + x + 3) + \frac{5}{x - 3}$$

- This means that $(x - 3)$ divides $(x^3 - 2x^2 - 4)$ a total of $(x^2 + x + 3)$ times with 5 leftover.

Exponent Rules

Definition

Let $x, y \in \mathbb{R}$ and $m, n \in \mathbb{N}$. Then

- $x^n = x \cdot x \cdots x$ (n times)
- $x^0 = 1$, for $x \neq 0$
- $x^{-n} = \dfrac{1}{x^n}$, for $x \neq 0$
- $x^{m/n} = \sqrt[n]{x^m}$ or $\left(\sqrt[n]{x}\right)^m$, for $x \geq 0$, where $\sqrt[n]{\ }$ denotes the nth root.

Theorem

Let $x, y, a, b \in \mathbb{R}$ with $x, y > 0$. Then

- $x^a x^b = x^{a+b}$
- $\dfrac{x^a}{x^b} = x^{a-b}$, for $x \neq 0$
- $(x^a)^b = x^{ab}$
- $(xy)^a = x^a y^a$, for $x \geq 0$, $y \geq 0$
- $\left(\dfrac{x}{y}\right)^a = \dfrac{x^a}{y^a}$, for $x \geq 0$, $y > 0$

Example

Assuming $x, y > 0$, simplify the following expression: $\dfrac{3x^{-2}y^3x}{y^2\sqrt{x}}$.

Write your answer in the form $\dfrac{ay^m}{b\sqrt{x^n}}$ for suitable $a, b, m, n \in \mathbb{N}$.

Solution: Using the Exponent Rules, we have:

$$
\begin{aligned}
\frac{3x^{-2}y^3x}{y^2\sqrt{x}} &= \frac{3x^{-2}y^3x}{y^2x^{\frac{1}{2}}}, && \text{since } \sqrt{x} = x^{\frac{1}{2}}, \\
&= \frac{3x^{-2}yx}{x^{\frac{1}{2}}}, && \text{since } \frac{y^3}{y^2} = y, \\
&= \frac{3y}{x^{\frac{3}{2}}}, && \text{since } \frac{x^{-2}x}{x^{\frac{1}{2}}} = \frac{x^{-1}}{x^{\frac{1}{2}}} = x^{-\frac{3}{2}} = \frac{1}{x^{\frac{3}{2}}}, \\
&= \frac{3y}{\sqrt{x^3}}, && \text{since } x^{\frac{3}{2}} = \sqrt{x^3}.
\end{aligned}
$$

Inequality Notation

- $a < b$ means a is to the $\boxed{\text{left}}$ of b (i.e., a is $\boxed{\text{strictly less}}$ than b)
- $a \leq b$ means a is to the $\boxed{\text{left of or the same}}$ as b (i.e., a is $\boxed{\text{less than or equal}}$ to b)
- $a > b$ means a is to the $\boxed{\text{right}}$ of b (i.e., a is $\boxed{\text{strictly greater}}$ than b)
- $a \geq b$ means a is to the $\boxed{\text{right of or the same}}$ as b (i.e., a is $\boxed{\text{greater than or equal}}$ to b)

Mnemonics: The $<$ symbol is like a slanted **L** and stands for "**L**ess than".

Also, the alligator method (ask a classmate).

Example

The following are true:

- $1 < 2$
- $-5 < -2$
- $1 \leq 2$
- $1 \leq 1$
- $4 \geq \pi > 3$
- $7.23 \geq -7.23$

Properties of Inequalities

The following theorem lists **Algebraic Rules for Inequalities**.

Theorem

Let a, b and c be nonzero real numbers.

(a) If $a < b$ and $c > 0$, then $ac < bc$.
   Multiplying by a positive number preserves the inequality.

(b) If $a < b$ and $c < 0$, then $ac > bc$.
   Multiplying by a negative number flips the inequality.

(c) If $0 < a < b$, then $\dfrac{1}{a} > \dfrac{1}{b}$.
   Can take reciprocal if both positive (but flip the inequality).

(d) If $a < b < 0$, then $\dfrac{1}{a} > \dfrac{1}{b}$.
   Can take reciprocal if both negative (but flip the inequality).

(e) If $a < b$, then $a + c < b + c$.
   Adding a number to both sides preserves the inequality.

- We only list the rules for $<$, but similar rules hold for each of $\leq$, $>$ and $\geq$.

Properties of Inequalities

Theorem

Let A and B be any real expressions. Then

(a) $AB > 0$ if and only if ($A > 0$ and $B > 0$) or ($A < 0$ and $B < 0$).

(b) $AB < 0$ if and only if ($A > 0$ and $B < 0$) or ($A < 0$ and $B > 0$).

Theorem

A polynomial can change sign only at values where it is equal to zero.

A quotient of polynomials can change sign only at values where its numerator or denominator is equal to zero.

Solving Basic Inequalities

Example

- Find all values of x satisfying $3x + 1 > 2x - 3$.
- Write your answer in both interval and set-builder notation.
- Draw a number line indicating your solution set.

Solution:

- Subtract $2x$ from both sides: $x + 1 > -3$.
- Subtract 1 from both sides: $x > -4$.
- The solution is the interval $(-4, \infty)$.
- We can draw the solution on the number line as follows:

- In set-builder notation the solution is the set $\{x \in \mathbb{R} \mid x > -4\}$.

Solving Basic Inequalities

Example

Solve the inequality $4 > 3x - 2 \geq 2x - 1$.

Solution:
We need both $4 > 3x - 2$ and $3x - 2 \geq 2x - 1$ to be true:

$$\begin{array}{ccc} 4 > 3x - 2 & \textbf{and} & 3x - 2 \geq 2x - 1 \\ 6 > 3x & \textbf{and} & x - 2 \geq -1 \\ 2 > x & \textbf{and} & x \geq 1 \\ x < 2 & \textbf{and} & x \geq 1 \end{array}$$

- What numbers satisfy both of these?
- We require $x \geq 1$ but also $x < 2$:
- This gives all the numbers between 1 and 2, including 1 but not including 2.
- That is, the solution set is the interval $[1, 2)$.
- In set-builder notation this is the set $\{x \in \mathbb{R} \mid 1 \leq x < 2\}$.

Solving Rational Inequalities using the Split Point Technique

The **Split Point Technique** is a method that can be used to solve rational inequalities.

It works because from a previous theorem (i.e., a quotient of polynomials can change sign only at values where its numerator or denominator is equal to zero).

(1) Move all terms to **one side** to set the inequality to 0.

(2) Combine terms using a **common denominator**.

(3) **Factor** the numerator and denominator.

(4) Identify **split points** (where the numerator or denominator equals 0).

(5) Draw a **number line**, mark split points with **open/closed circles** based on the inequality.

(6) These points split the number line up into subintervals. Pick a **test point** for each subinterval to test if the expression is positive or negative.

(7) Write the solution in set-builder notation, using $\cup$ if necessary.

Example

Write the solution set to the following inequality using intervals: $\dfrac{2-x}{2+x} \geq 1$.

Solution:

- If we multiply both sides by $2+x$ we need to do **case work**:
  - **Why?** Because x is a variable ands we don't know if $2+x$ is positive or negative!
- Instead, we follow the steps for the split point technique:

Start with original problem: $$\frac{2-x}{2+x} \geq 1$$

Move to one side: $$\frac{2-x}{2+x} - 1 \geq 0$$

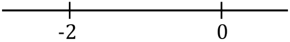
Common denominator: $$\frac{2-x}{2+x} - \frac{2+x}{2+x} \geq 0$$

Combine fractions: $$\frac{(2-x)-(2+x)}{2+x} \geq 0$$

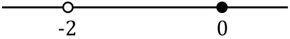
Expand numerator: $$\frac{2-x-2-x}{2+x} \geq 0$$

Simplify numerator: $$\frac{-2x}{2+x} \geq 0$$

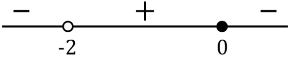
- We want to solve: $\dfrac{-2x}{2+x} \geq 0$
- Split points are $x = 0$ (makes the numerator 0) and $x = -2$ (makes the denominator 0)



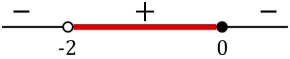
- Point $x = 0$ is included since if we sub in $x = 0$ we get $0 \geq 0$ which is true.
- Point $x = -2$ is not included since we cannot divide by zero.
- Indicate this with open/closed circles on the number line (open means omit):



- Choose a test point from each of the three subintervals and sub into $\dfrac{-2x}{2+x}$.
- When $x = -3$ it is negative. When $x = -1$ it is positive. When $x = 1$ it is negative.



- Since we want $\dfrac{-2x}{2+x} \geq 0$, we look at where the $+$ signs are and closed circles.



- Since there is a closed circle at 0, we include it. Thus, the solution is $(-2, 0]$.

Example

Write the solution set to the inequality using intervals: $\dfrac{2}{x+2} > 3x+3$.

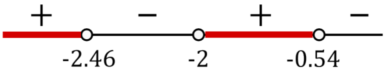
Solution: Using the split point technique gives:

$$
\begin{aligned}
\text{Start with original problem:} \quad & \frac{2}{x+2} > 3x+3 \\
\text{Move to one side:} \quad & \frac{2}{x+2} - (3x+3) > 0 \\
\text{Common denominator:} \quad & \frac{2}{x+2} - \frac{(3x+3)(x+2)}{x+2} > 0 \\
\text{Combine fractions:} \quad & \frac{2-(3x+3)(x+2)}{x+2} > 0 \\
\text{Expand numerator:} \quad & \frac{2-3x^2-6x-3x-6}{x+2} > 0 \\
\text{Simplify numerator:} \quad & \frac{-(3x^2+9x+4)}{x+2} > 0
\end{aligned}
$$

- We want to solve: $\dfrac{-(3x^2+9x+4)}{x+2} > 0$
- The denominator is zero when $x = -2$.
- Using the quadratic formula, the numerator is zero when

$$x = \frac{-9 \pm \sqrt{33}}{6}$$

  (these two numbers are approximately -2.46 and -0.54).
- Since the inequality uses "$>$" and $0 > 0$ is false, we do not include the split points.
- After choosing suitable test points and determining the sign of $\dfrac{-(3x^2+9x+4)}{x+2}$ we have



- Looking where the $+$ symbols are located gives the solution:

$$\left(-\infty, \frac{-9-\sqrt{33}}{6}\right) \bigcup \left(-2, \frac{-9+\sqrt{33}}{6}\right).$$

- When writing the final answer we use **exact** expressions for numbers in mathematics, not approximations (unless stated otherwise).

Absolute Values

Definition: Absolute Value

The $\boxed{\text{absolute value}}$ of a real number a is $|a| = \begin{cases} a, & \text{if } a \geq 0, \\ -a, & \text{if } a < 0. \end{cases}$

- The absolute value does **not** turn all minus signs into plus signs:

$$\left|3a - 6\right| \neq 3a + 6 \qquad \text{(unless } a = 0\text{)}.$$

- Instead, we write

$$\left|3a - 6\right| = \begin{cases} 3a - 6, & \text{if } 3a - 6 \geq 0, \\ -(3a - 6), & \text{if } 3a - 6 < 0, \end{cases} \qquad = \begin{cases} 3a - 6, & \text{if } a \geq 2, \\ 6 - 3a, & \text{if } a < 2. \end{cases}$$

- For $a \in \mathbb{R}$, the absolute value $|a|$ is the **distance** between a and 0 on the real number line.

Definition: Distance

Let $a, b \in \mathbb{R}$. The $\boxed{\text{distance}}$ between a and b is $\operatorname{dist}(a, b) = \left|b - a\right|$.

Properties of Absolute Values

Theorem

Let $x, y \in \mathbb{R}$. Then

(i) $\sqrt{x^2} = |x|$

(ii) $|x| \geq 0$

(iii) $|-x| = |x|$

(iv) $-|x| \leq x \leq |x|$

(v) $|xy| = |x|\,|y|$

(vi) $|x + y| \leq |x| + |y|$ (the **triangle inequality**)

(vii) $|x - y| \geq \big||x| - |y|\big|$ (the **reverse triangle inequality**)

Proof.
Try to prove this at home using the definition of absolute value.

Absolute Values and Inequalities

- We can solve inequalities with absolute values by first getting rid of the absolute value.
- **Case 1:** $a > 0$
  - $|x| = a$ has solutions $x = \pm a$.
  - $|x| \leq a$ means $x \geq -a$ **and** $x \leq a$ (that is, $-a \leq x \leq a$).
  - $|x| < a$ means $x < -a$ **and** $x < a$ (that is, $-a < x < a$).
  - $|x| \geq a$ means $x \leq -a$ **or** $x \geq a$.
  - $|x| > a$ means $x < -a$ **or** $x > a$.
- **Case 2:** $a < 0$
  - $|x| = a$ has no solutions.
  - Both $|x| \leq a$ and $|x| < a$ have no solutions.
  - Both $|x| \geq a$ and $|x| > a$ have solution set $\mathbb{R}$.
- **Case 3:** $a = 0$
  - $|x| = 0$ has solution $x = 0$.
  - $|x| < 0$ has no solutions.
  - $|x| \leq 0$ has solution $x = 0$.
  - $|x| > 0$ has solution set $\{x \in \mathbb{R} \,|\, x \neq 0\}$.
  - $|x| \geq 0$ has solution set $\mathbb{R}$.

Example

Solve $|x-5|<7$.

Solution:

- Recall that when $a>0$, then
$$|z|<a \qquad \Longrightarrow \qquad -a<z<a.$$
- Thus, we have
$$-7<x-5<7.$$
- Adding 5 everywhere gives
$$-2<x<12.$$
- Therefore the solution is the interval $(-2,12)$.

Example

Solve $|2 - z| \leq 7$.

Solution:

- Recall that when $a > 0$,
$$|z| < a \qquad \Longrightarrow \qquad -a < z < a.$$
- Thus, we have
$$-7 \leq 2 - z \leq 7.$$
- Subtracting 2 everywhere gives
$$-9 \leq -z \leq 5.$$
- Now multiplying by -1 gives
$$9 \geq z \geq -5.$$
Remember to reverse the inequality sign!
- We can rearrange this as
$$-5 \leq z \leq 9.$$
- Therefore the solution is the interval $[-5, 9]$.

Example

Solve $|2 - x| \geq 7$.

Solution:

- Recall that when $a > 0$,

$$|z| \geq a \qquad \Longrightarrow \qquad z \leq -a \ \textbf{or} \ z \geq a.$$

- Therefore, we have, either:

$$2 - x \leq -7 \quad \textbf{or} \quad 2 - x \geq 7.$$

$$9 \leq x \quad \textbf{or} \quad -5 \geq x.$$

$$x \geq 9 \quad \textbf{or} \quad x \leq -5.$$

- In interval notation, either x is in $[9, \infty)$ **or** x is in $(-\infty, -5]$.
- Therefore, the solution is $(-\infty, -5] \cup [9, \infty)$.

Example

Solve $0 < |x-5| \leq 7$.

Solution:

- We first solve both inequalities separately.

  (1) $0 < |x-5|$:
  - This is the same as $|x-5| > 0$.
  - Since $|z| > 0$ for all $z \in \mathbb{R}$ except when $z = 0$, the solutions to $|x-5| > 0$ are all real numbers except 5, that is, $x \neq 5$.
  - In interval notation we write this as $(-\infty, 5) \cup (5, \infty)$.

  (2) $|x-5| \leq 7$:
  - This is the equivalent to
  $$-7 \leq x - 5 \leq 7.$$
  - Adding 5 everywhere gives
  $$-2 \leq x \leq 12.$$
  - Therefore, the solution to $|x-5| \leq 7$ is the interval $[-2, 12]$.

- To combine **(1)** and **(2)** we need combine $x \neq 5$ with $x \in [-2, 12]$.

- Omitting 5 from the interval $[-2, 12]$ gives the solution
$$[-2, 5) \cup (5, 12].$$

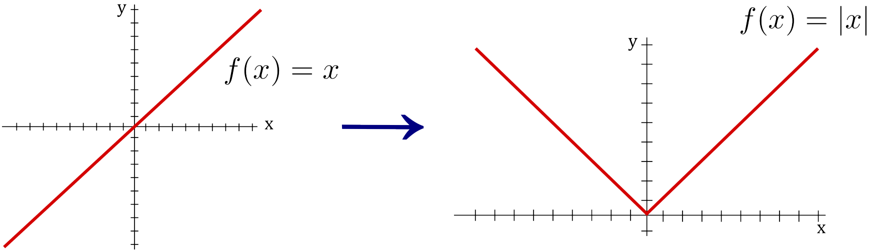
The Absolute Value Function

Definition

The $\boxed{\textbf{absolute value function}}$ is defined by:

$$|x| = \begin{cases} x, & \text{if } x \geq 0, \\ -,x & \text{if } x < 0. \end{cases}$$

To visualize, take the straight line $f(x) = x$ and **“reflect”** the negative part to make it positive:

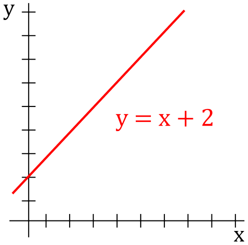


Lines

- Let $m, b \in \mathbb{R}$ be fixed real numbers.
- A $\boxed{\textbf{linear function}}$ is any function of the form $y = mx + b$.
- Visually, a linear function is a (non-vertical) **straight line** in the xy-plane.
- Vertical lines cannot represent the graph of a function (as we review later).

Example

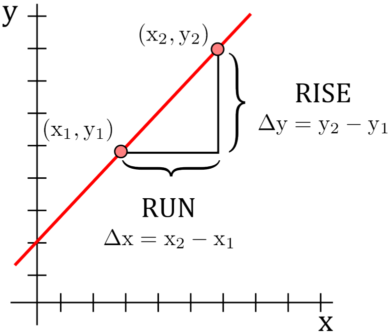
The linear function $y = x + 2$ is graphed below.



Slopes of Lines

- The **steepness** of a line is called the **slope** of the line.
- We typically use the letter **m** to represent the slope.
- Let (x_1, y_1) and (x_2, y_2) be two points that lie on a line.
- Then the **slope** of a (nonvertical) line through (x_1, y_1) and (x_2, y_2) is:

$$m = \frac{\text{rise}}{\text{run}} = \frac{\text{change in } y}{\text{change in } x} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\Delta y}{\Delta x}.$$

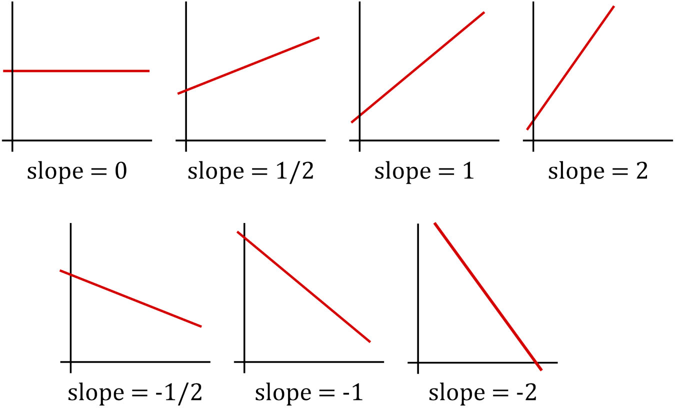


If the two points form a vertical line then $x_2 - x_1 = 0$.

- In this case, we say the slope is not defined.

Slopes

Below are visual representations of lines with a zero slope, positive slope and negative slope:



x and y-intercepts

- **x-intercepts** are where a graph **crosses the x-axis**.
- More specifically, an **x-intercept** is a point in the equation where the **y-value is zero**.
- **y-intercepts** are where the graph **crosses the y-axis**.
- More specifically, a **y-intercept** is a point in the equation where the **x-value is zero**.

Example

Assume $m \neq 0$. Find the x and y-intercepts of $y = mx + b$.

Solution:

- To obtain x-intercept(s), set $y = 0$ and solve for x:

$$0 = mx + b \qquad \to \qquad mx = -b \qquad \to \qquad x = -\frac{b}{m}.$$

- To obtain y-intercept(s), set $x = 0$ and solve for y:

$$y = m(0) + b \qquad \to \qquad y = b.$$

- Therefore, $y = mx + b$ has an x-intercept of $x = -b/m$ and y-intercept of $y = b$.

Equations for Lines

There are **three** common forms for the equation of a line:

Point-Slope Form
An equation of a line passing through (x_1, y_1) and having slope m is:

$$y - y_1 = m(x - x_1).$$

Slope-Intercept Form
An equation of a line with slope m and y-intercept b is:

$$y = mx + b.$$

General Form
We can write the equation of any line in the form:

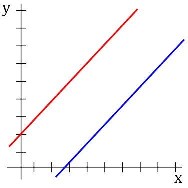
$$Ax + By + C = 0,$$

where A, B, C are constants and A and B are not both 0.

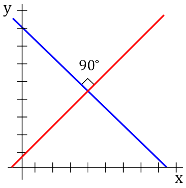
Parallel and perpendicular lines

- It is often necessary to know if two lines are parallel or perpendicular.
- Let m_1 and m_2 be the slopes of the nonvertical lines L_1 and L_2.
  - L_1 and L_2 are **parallel** if and only if $m_1 = m_2$.
  - L_1 and L_2 are **perpendicular** if and only if $m_2 = \dfrac{-1}{m_1}$.
- In the case of perpendicular lines, we say their slopes are **negative reciprocals**.

Parallel Lines



Perpendicular Lines



Example

- The lines $y = 2x - 1$ and $y = 2x + 8$ are **parallel**.
- The lines $y = 2x - 1$ and $y = -\frac{1}{2}x + 3$ are **perpendicular**.
- The lines $y = 2x - 1$ and $y = 3x + 2$ are **neither** parallel nor perpendicular.

Example

Find an equation of the line through the two points $(0, 3)$ and $(-2, 4)$.

Solution:

- We use the **slope formula** on $(x_1, y_1) = (0, 3)$ and $(x_2, y_2) = (-2, 4)$ to find m:

$$m = \frac{(4) - (3)}{(-2) - (0)} = \frac{1}{-2} = -\frac{1}{2}.$$

- Now using the **point-slope formula** we get an equation to be:

$$y - 3 = -\frac{1}{2}(x - 0) \quad \rightarrow \quad y = -\frac{1}{2}x + 3.$$

Example

Find an equation of the line with slope 7 and through point $(1, -2)$.

Solution:

- Using the **point-slope formula** with $m = 7$ and $(x_1, y_1) = (1, -2)$ gives:

$$y - (-2) = 7(x - 1)$$

- Therefore,

$$y = 7x - 9.$$

Example

Find an equation of the line with slope 2 and y-intercept 4.

Solution:

- Using the **slope-intercept formula** with $m = 2$ and $b = 4$ we get

$$y = 2x + 4$$

Example

Find an equation of the line through point $(5, 3)$ and parallel to the line $2x + 4y + 2 = 0$.

Solution:

- The line $2x + 4y + 2 = 0$ can be written as:

$$4y = -2x - 2$$

- Writing this in **slope-intercept** form gives:

$$y = -\frac{1}{2}x - \frac{1}{2}.$$

- This line has slope $-1/2$.
- Since our line is **parallel** to it, we have $m = -1/2$.
- Now we have a point $(x_1, y_1) = (5, 3)$ and slope $m = -1/2$.
- The **point-slope formula** gives:

$$y - 3 = -\frac{1}{2}(x - 5)$$

$$y = -\frac{1}{2}x + \frac{11}{2}.$$

Example

Find an equation of the line with y-intercept 4 and perpendicular to the line $y = -\frac{2}{3}x + 3$.

Solution:

- The line $y = -\frac{2}{3}x + 3$ has slope $-2/3$.
- Since our line is perpendicular to it, the slope of our line is the **negative reciprocal**, hence:

$$m = 3/2$$

- Now we have $b = 4$ (given in the question) and $m = 3/2$.
- Thus by the **slope-intercept formula**, an equation of the line is

$$y = \frac{3}{2}x + 4.$$

Example

Are the two lines $7x + 2y + 3 = 0$ and $6x - 4y + 2 = 0$ perpendicular? Are they parallel? If they are not parallel, what is their point of intersection?

Solution:

- The first line L_1 is:

$$7x + 2y + 3 = 0 \quad \rightarrow \quad 2y = -7x - 3 \quad \rightarrow \quad y = -(7/2)x - 3.$$

- It has slope $m_1 = -7/2$.
- The second line L_2 is:

$$6x - 4y + 2 = 0 \quad \rightarrow \quad -4y = -6x - 2 \quad \rightarrow \quad y = \frac{3}{2}x + \frac{1}{2}.$$

- It has slope $m_2 = 3/2$.
- Since $m_1 \cdot m_2 \neq -1$ (they are not negative reciprocals), the lines are not perpendicular.
- Since $m_1 \neq m_2$ the lines are not parallel.

Homework: Find the points of intersection by setting y-values equal.

Conics

- We next review equations of parabolas, circles, ellipses and hyperbolas.
- Each equation presented is the standard form for the type of conic.

Mnemonic:

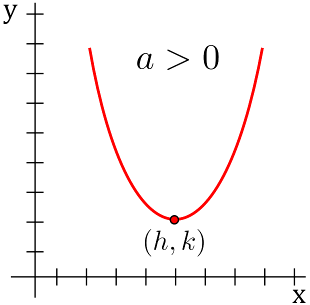
In each conic formula presented, the terms '**x-h**' and '**y-k**' will always appear.

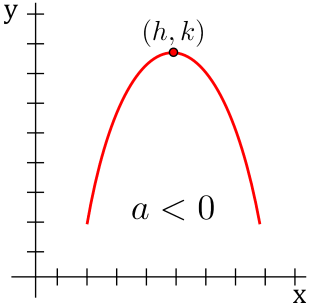
The point **(h,k)** will alway represent either the $\boxed{\textbf{centre}}$ or $\boxed{\textbf{vertex}}$ of the particular conic.

Vertical Parabola

The equation of a vertical parabola is:

$$\boxed{y - k = a(x - h)^2}$$



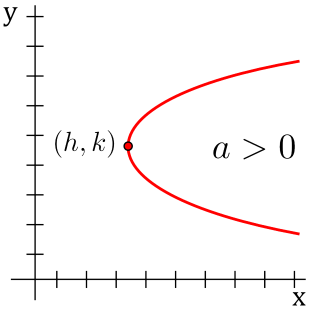


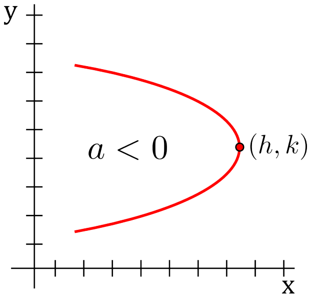
- (h, k) is the **vertex** of the parabola.
- a is the vertical **stretch factor**.
- If $a > 0$, the parabola opens **upward**.
- If $a < 0$, the parabola opens **downward**.

Horizontal Parabola

The equation of a horizontal parabola is:

$$\boxed{x - h = a(y-k)^2}$$



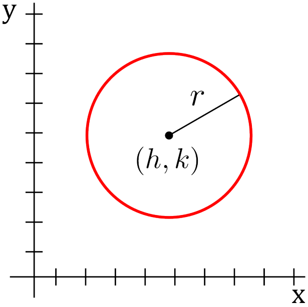


- (h, k) is the **vertex** of the parabola.
- a is the horizontal **stretch factor**.
- If $a > 0$, the parabola opens **right**.
- If $a < 0$, the parabola opens **left**.

Circle

The equation of a circle is:

$$\boxed{(x-h)^2 + (y-k)^2 = r^2}$$

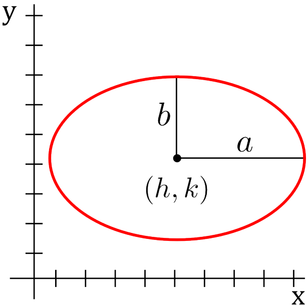


- (h, k) is the **centre** of the circle.
- r is the **radius** of the circle.

Ellipse

The equation of an ellipse is:

$$\boxed{\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1}$$

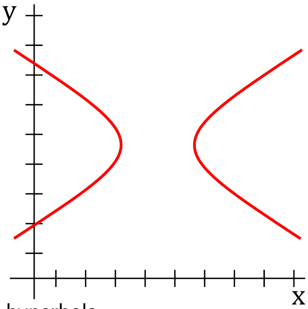


- (h, k) is the **centre** of the ellipse.
- a is the **horizontal distance** from the centre to the edge of the ellipse. b is the **vertical distance** from the centre to the edge.

Horizontal Hyperbola

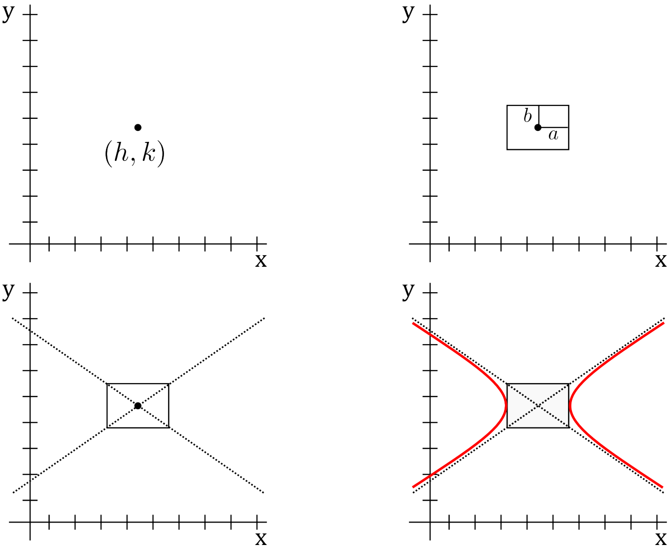
The equation of a horizontal hyperbola is:

$$\boxed{\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1}$$



- (h, k) is the **centre** of the hyperbola.
- a, b are the **reference box** values. The box has a centre of (h, k).
- a is the **horizontal distance** from the centre to the edge of the box.
- b is the **vertical distance** from the centre to the edge of the box.

Sketching a Horizontal Hyperbola

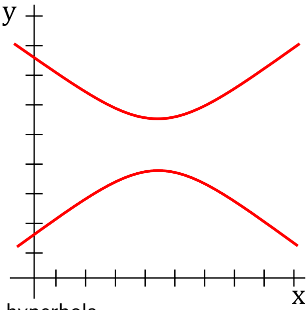


The dotted lines above are called the **asymptotes** of the hyperbola.

Vertical Hyperbola

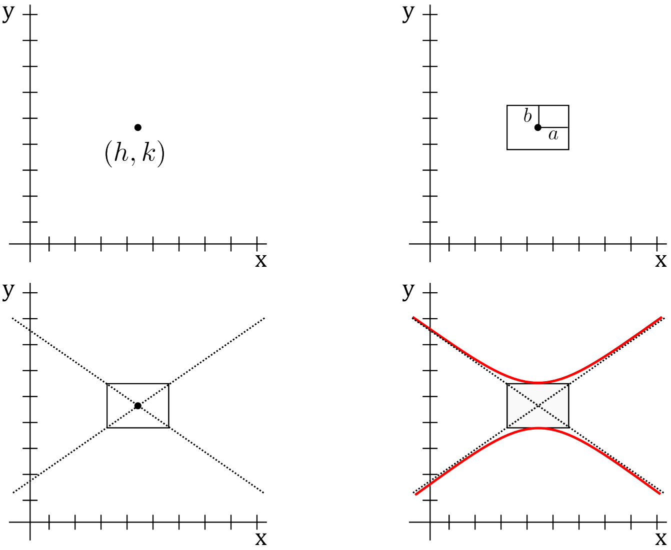
The equation of a vertical hyperbola is:

$$\boxed{\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = -1}$$



- (h, k) is the **centre** of the hyperbola.
- a, b are the **reference box** values. The box has a centre of (h, k).
- a is the **horizontal distance** from the centre to the edge of the box.
- b is the **vertical distance** from the centre to the edge of the box.

Sketching a Vertical Hyperbola



The dotted lines above are called the **asymptotes** of the hyperbola.

Completing the Square

- **Completing the square** is an important and useful technique.
- It allows us to determine the center of a circle/ellipse or the vertex of a parabola.
- It is also used to derive the quadratic formula (or solve quadratic equations).
- The main idea behind completing the square is to turn:

$$ax^2 + bx + c$$

into

$$a(x-h)^2 + k.$$

- One way to complete the square is to use the following formula:

$$ax^2 + bx + c = a\left(x + \frac{b}{2a}\right)^2 - \frac{b^2}{4a^2} + c.$$

- As this formula is complicated, we outline some easier steps in the next example which use the factoring formula for **perfect square trinomials**:

$$r^2 + 2rs + s^2 = (r+s)^2$$

$$r^2 - 2rs + s^2 = (r-s)^2$$

Solving a Quadratic by Completing the Square

Example

Solve $2x^2 + 12x - 32 = 0$ by completing the square.

Solution: Because I want to demonstrate what you should do when the a value (in front of x^2) is not 1, we will **not** divide by 2 first (though usually you would).

$2x^2 + 12x - 32 = 0$	Start with original equation.
$2x^2 + 12x = 32$	Move the number over to the other side.
$2(x^2 + 6x) = 32$	Factor out the "a" from the $ax^2 + bx$ expression.
$6 \rightarrow \frac{6}{2} = 3 \rightarrow 3^2 = \mathbf{9}$	Take # in front of x, **divide** by 2, **square** it.
$\mathbf{2}(x^2 + 6x + \mathbf{9}) = 32 + \mathbf{2} \cdot \mathbf{9}$	"Add" to both sides (remember the a value!)
$2(x+3)^2 = 50$	Factor the resulting perfect square trinomial.
	You have now completed the square!!
$x = 2$ or $x = -8$	Divide by 2, take square roots, subtract 3.

Example

Find the center and radius of the circle:

$$y^2 + x^2 - 12x + 8y + 43 = 0.$$

Solution:

- We need to complete the square **twice**!
- Collect the terms with x together and the terms with y together:

$$(x^2 - 12x) + (y^2 + 8y) = -43$$

- Add **36** to both sides for the x term ($-12 \to -6 \to 36$).
- Add **16** to both sides for the y term ($8 \to 4 \to 16$):

$$(x^2 - 12x + \mathbf{36}) + (y^2 + 8y + \mathbf{16}) = -43 + \mathbf{36} + \mathbf{16}$$

- Factoring gives:

$$(x-6)^2 + (y+4)^2 = 3^2.$$

- Therefore, the **centre** of the circle is $(6, -4)$ and the **radius** is 3.

Determining the type of conic

- An equation of the form

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$$

gives rise to a graph that can be generated by performing a conic section.

- These are parabolas, circles, ellipses, and hyperbolas.
- The Bxy term involves conic rotation.
- For simplicity, we will **omit** the Bxy term.
- To determine the type of graph we analyze the values of A and C.
  - If $A = C$, the graph is a **circle**.
  - If $AC > 0$ (and $A \neq C$), the graph is an **ellipse**.
  - If $AC = 0$, the graph is a **parabola**.
  - If $AC < 0$, the graph is a **hyperbola**.

Example

What type of conic is $4x^2 - y^2 - 8x + 8 = 0$? Write it in standard form.

Solution:

- We see that $A = 4$ and $C = -1$.
- Since $AC < 0$, the conic is a hyperbola.
- We will complete the square. First collect x's and y's:

$$(4x^2 - 8x) - y^2 = -8$$

- Now we factor out 4 from the x terms.

$$4(x^2 - 2x) - y^2 = -8$$

- Notice that we don't need to complete the square for the y terms.
- For the x terms, we add **1** (divide -2 by two and square it).
- Taking into consideration that the "a" value is 4:

$$\mathbf{4}(x^2 - 2x + \mathbf{1}) - y^2 = -8 + \mathbf{4} \cdot \mathbf{1}$$

- Factoring gives:

$$4(x-1)^2 - y^2 = -4$$

- A hyperbola in standard form has ± 1 on the right side and a positive x^2 on the left side.
- Thus, we must divide by 4:

$$(x-1)^2 - \frac{y^2}{4} = -1$$

- Now we can see that the equation represents a **vertical hyperbola** with centre $(1, 0)$.

Trigonometry - Angles (Degrees and Radians)

- Units of angular measurement include degrees, radians, grads, and turns.
- Mathematicians tend to deal with $\boxed{\textbf{radians}}$.
- Formulas in calculus are more elegant when using radians (rather than degrees).
- The relationship between degrees and radians is:

$$\pi \textbf{ radians} = 180^\circ$$

- Thus, π radians is half a turn around a circle and 2π radians is a full turn around a circle.
- Using this formula, some common angles can be derived:

Degrees	0°	30°	45°	60°	90°	120°	135°	150°	180°	270°	360°
Radians	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{3\pi}{4}$	$\frac{5\pi}{6}$	π	$\frac{3\pi}{2}$	2π

Example

Convert 45° to radians.

Solution:

To convert 45° to radians, multiply by $\frac{\pi}{180^\circ}$ to get $\frac{\pi}{4}$.

Example

Convert $\frac{5\pi}{6}$ radians to degrees.

Solution:

To convert $\frac{5\pi}{6}$ radians to degrees, multiply by $\frac{180^\circ}{\pi}$ to get 150°.

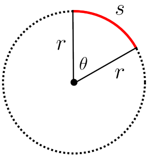
Mnemonic: How do you know if you multiply by $\frac{\pi}{180^\circ}$ or $\frac{180^\circ}{\pi}$?

- In the first example, the degree symbols canceled giving us a unit-less answer (radians).
- In the second example, the degree symbol was on top which gave us an answer in degrees.

From now on, we will only use **radian measure**, unless specified otherwise.

Sectors of Circles

- The figure below illustrates a sector of a circle with **central angle** θ and radius r **subtending** an arc with **length** s.



- When θ is measured in radians, we have the following formulas: $\theta = \dfrac{s}{r}$ or $s = r\theta$.
- The area of the sector is equal to: **Sector Area** $= \dfrac{1}{2}r^2\theta$.

Example

If a circle has radius 3 cm, then an angle of 2 rad is subtended by an arc of 6 cm.

Example

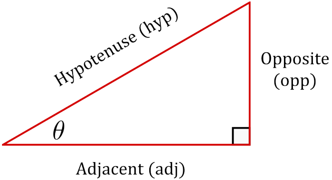
If we substitute $\theta = 2\pi$ into the sector area formula we get the area of a circle:

$$A = \frac{1}{2}r^2(2\pi) = \pi r^2.$$

Defining Trigonometric Functions

- $\boxed{\textbf{Trigonometric functions}}$ (also called **circular functions**) are functions of an angle.
- They relate angles of a triangle to the lengths of its sides.
- There are six basic trigonometric functions:
  - Sine (abbreviated by sin)
  - Cosine (abbreviated by cos)
  - Tangent (abbreviated by tan)
  - Cosecant (abbreviated by csc)
  - Secant (abbreviated by sec)
  - Cotangent (abbreviated by cot)
- We first define trigonometric functions in terms of ratios of two sides of a **right angle triangle** containing the angle.
- We then look at definitions of trigonometric functions in terms of the **unit circle**.

Defining Trigonometric Functions



- With reference to the above triangle, for an acute angle θ (that is, $0 \leq \theta < \pi/2$), the six trigonometric functions can be described as follows:

$$\sin\theta = \frac{\text{opp}}{\text{hyp}} \qquad \csc\theta = \frac{\text{hyp}}{\text{opp}}$$

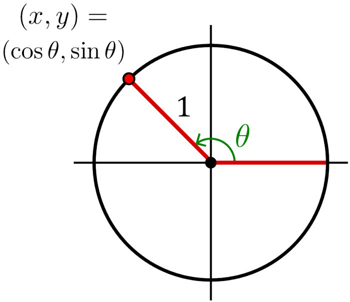
$$\cos\theta = \frac{\text{adj}}{\text{hyp}} \qquad \sec\theta = \frac{\text{hyp}}{\text{adj}}$$

$$\tan\theta = \frac{\text{opp}}{\text{adj}} \qquad \cot\theta = \frac{\text{adj}}{\text{opp}}$$

- **Mnemonic:** SOH CAH TOA
- The above description does not apply to **obtuse** or **negative angles**.

Formal Definition

- We first define sine and cosine using a **unit** circle.
- Then define the other trigonometric functions in terms of sine and cosine.
- Take a line originating at the origin (making an angle of θ with the positive half of the x-axis) and suppose this line intersects the unit circle at the point (x, y).
- **Definition:**
  The x- and y-coordinates of this point of intersection are equal to $\cos\theta$ and $\sin\theta$.



- Positive angles rotate **counter-clockwise** around the circle.
- Negative angles rotate **clockwise** around the circle.
- For angles greater than 2π simply rotate around the circle multiple times.
- In this way, sine and cosine become **periodic functions** with period 2π:

$$\sin\theta = \sin(\theta + 2\pi k) \qquad\qquad \cos\theta = \cos(\theta + 2\pi k)$$

  for any angle θ and any integer k.

- The other trigonometric functions are defined in terms of sine and cosine:

$$\tan\theta = \frac{\sin\theta}{\cos\theta}$$

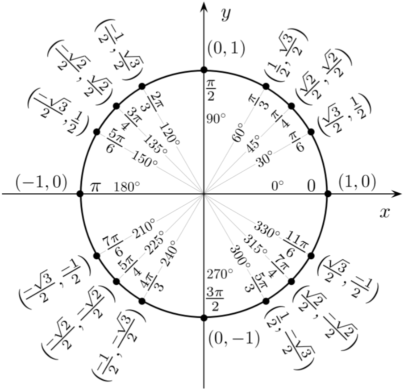
$$\sec\theta = \frac{1}{\cos\theta}$$

$$\csc\theta = \frac{1}{\sin\theta}$$

$$\cot\theta = \frac{\cos\theta}{\sin\theta}$$

Unit Circle

- The **unit circle** depicts **exact** values of trigonometric functions for some angles.
- Remember that the x-coordinate is $\cos\theta$ and the y-coordinate is $\sin\theta$.

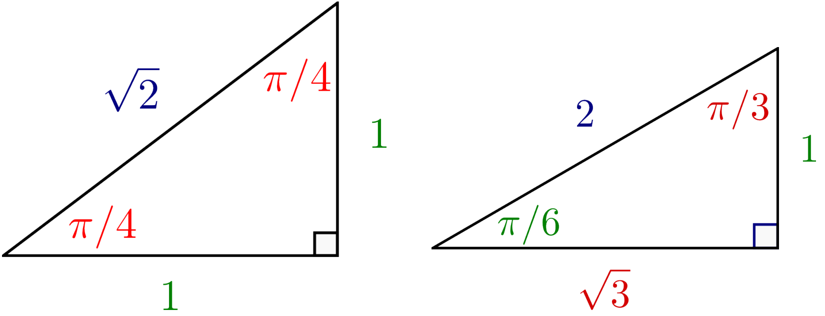


Example

From the unit circle we see that $\cos 5\pi/6 = -\sqrt{3}/2$ and $\sin 5\pi/6 = 1/2$.

Another calculation method uses **special triangles** and the **CAST rule**.

Special Triangles

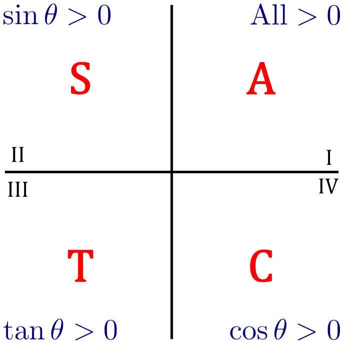


Mnemonic:

- The first triangle should be easy to remember.
- To remember the second triangle:
  - place the largest number (2) across from the largest angle ($90^\circ = \pi/2$).
  - Place the smallest number (1) across from the smallest angle ($30^\circ = \pi/6$).
  - Place the middle number ($\sqrt{3} \approx 1.73$) across from the middle angle ($60^\circ = \pi/3$).
- Double check using the Pythagorean Theorem that the sides satisfy $a^2 + b^2 = c^2$.

CAST Rule

- To determine the sign of sine, cosine and tangent we use the **CAST rule**.



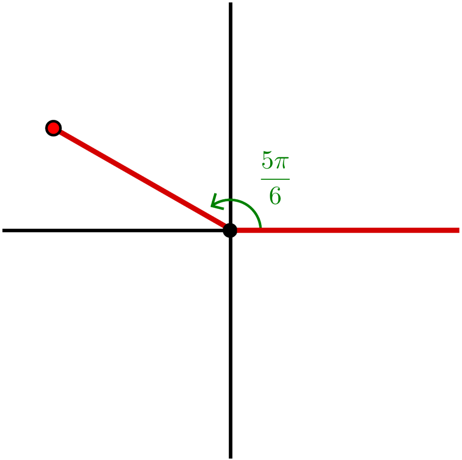
- In quadrant I, all three of $\sin\theta$, $\cos\theta$, $\tan\theta$ are positive.
- In quadrant II, only $\sin\theta$ is positive, while $\cos\theta$, $\tan\theta$ are negative.
- In quadrant III, only $\tan\theta$ is positive, while $\sin\theta$, $\cos\theta$ are negative.
- In quadrant IV, only $\cos\theta$ is positive, while $\sin\theta$, $\tan\theta$ are negative.
- **Mnemonic:** The **CAST** rule is already a mnemonic!
  Can also use: "**A**ll **S**tudents **T**ake **C**alculus".

Example

Determine $\sin 5\pi/6$, $\cos 5\pi/6$, $\tan 5\pi/6$, $\sec 5\pi/6$, $\csc 5\pi/6$ and $\cot 5\pi/6$ exactly by using the special triangles and CAST rule. Verify your answer with the unit circle.

Solution:

- Draw "xy-plane" and indicate our angle of $5\pi/6$ (**positive angles rotate counterclockwise**).



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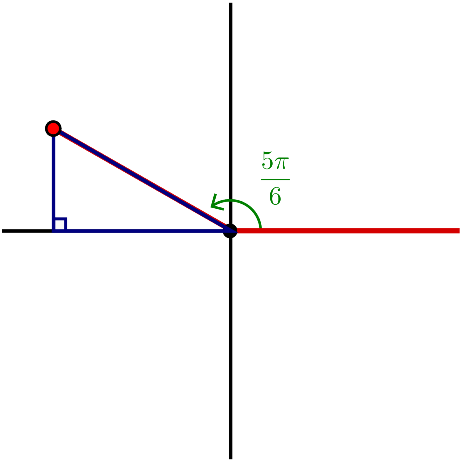
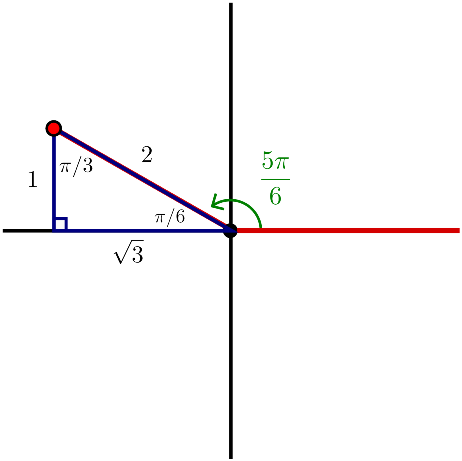
Next, we drop a perpendicular to the *x*-axis (never drop it to the *y*-axis!).



Continued...

- Notice that we can now figure out the angles in the triangle.
- Since $180^\circ = \pi$, we have an interior angle of $\pi - 5\pi/6 = \pi/6$ inside the triangle.
- As the **angles of a triangle add up to** $180^\circ = \pi$, the other angle must be $\pi/3$.
- This gives one of our special triangles. Labelling it accordingly:



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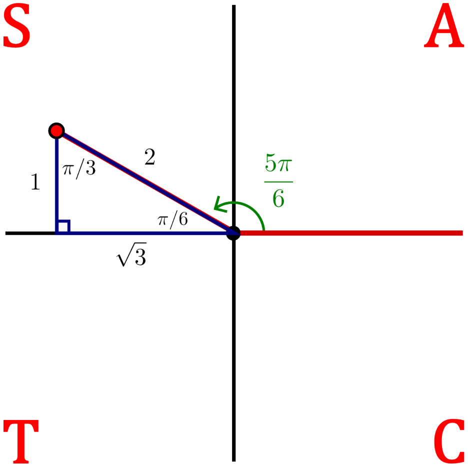
Now add the CAST rule to the diagram:



Continued...

- From the previous figure we see that $5\pi/6$ lies in quadrant II.
- By the CAST Rule, $\sin\theta$ is positive while $\cos\theta$ and $\tan\theta$ are negative.
- This gives us the **sign** of $\sin\theta$, $\cos\theta$ and $\tan\theta$.
- To determine the "**value**" use the triangle and SOH CAH TOA.
- Using $\sin\theta = opp/hyp$ we find a value of $1/2$.
  - But $\sin\theta$ is **positive** in quadrant II, therefore, $\sin\dfrac{5\pi}{6} = +\dfrac{1}{2}$.
- Using $\cos\theta = adj/hyp$ we find a value of $\sqrt{3}/2$.
  - But $\cos\theta$ is **negative** in quadrant II, therefore, $\cos\dfrac{5\pi}{6} = -\dfrac{\sqrt{3}}{2}$.
- Using $\tan\theta = opp/adj$ we find a value of $1/\sqrt{3}$.
  - But $\tan\theta$ is **negative** in quadrant II, therefore, $\tan\dfrac{5\pi}{6} = -\dfrac{1}{\sqrt{3}}$.
- To determine sec, csc and cot we use the definitions:

$$\csc\frac{5\pi}{6} = \frac{1}{\sin\frac{5\pi}{6}} = +2$$

$$\sec\frac{5\pi}{6} = \frac{1}{\cos\frac{5\pi}{6}} = -\frac{2}{\sqrt{3}}$$

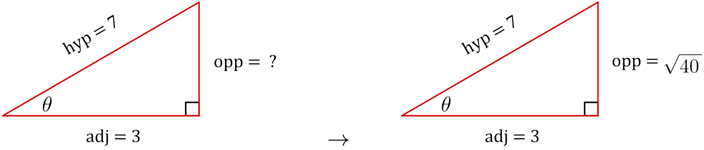
$$\cot\frac{5\pi}{6} = \frac{1}{\tan\frac{5\pi}{6}} = -\sqrt{3}$$

Example

If $\cos\theta = 3/7$ and $3\pi/2 < \theta < 2\pi$, then find $\cot\theta$.

Solution:

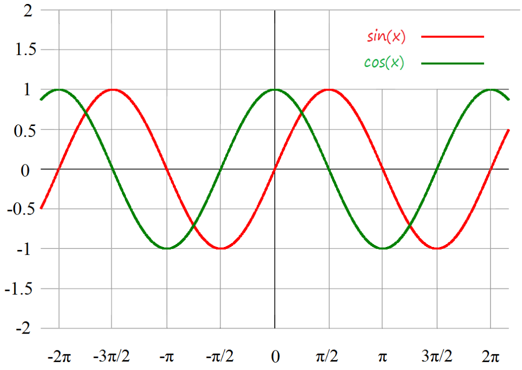
- We first draw a right angle triangle.
- Since $\cos\theta = \dfrac{adj}{hyp} = \dfrac{3}{7}$, we let the adjacent side have length 3 and the hypotenuse 7:



- Using the Pythagorean Theorem, we have $3^2 + (opp)^2 = 7^2 \;\rightarrow\;$ opp $= \sqrt{40}$.
- To find $\cot\theta$ we use the definition: $\cot\theta = \dfrac{1}{\tan\theta}$.
- Since we are given $3\pi/2 < \theta < 2\pi$, we are in the **fourth quadrant**.
- By the **CAST rule**, $\tan\theta$ is negative in this quadrant.
- As $\tan\theta = opp/adj$, we get:

$$\tan\theta = -\frac{\sqrt{40}}{3} \quad\rightarrow\quad \cot\theta = -\frac{3}{\sqrt{40}}.$$

- The graph of the functions $\sin x$ and $\cos x$ can be visually represented as:



- Both $\sin x$ and $\cos x$ have **domain** $(-\infty, \infty)$ and **range** $[-1, 1]$:

$$-1 \leq \sin x \leq 1 \qquad -1 \leq \cos x \leq 1.$$

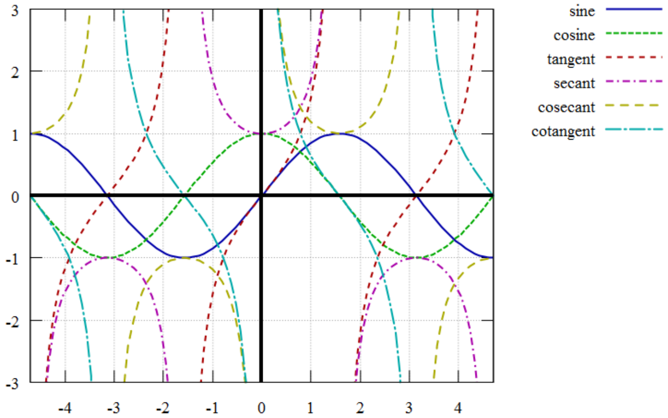
- The zeros of $\sin x$ occur at the integer multiples of π:

$$\sin x = 0 \text{ whenever } x = n\pi, \text{ where } n \in \mathbb{Z}.$$

- Similarly for $\cos x$ we have:

$$\cos x = 0 \text{ whenever } x = \pi/2 + n\pi, \text{ where } n \in \mathbb{Z}.$$

- The six basic trigonometric functions can be visually represented as:



- Both tangent and cotangent have range $(-\infty, \infty)$.
- Cosecant and secant have range $(-\infty, -1] \cup [1, \infty)$.
- Each of these functions is periodic.
- Tangent and cotangent have period π.
- Sine, cosine, cosecant and secant have period 2π.

Trigonometric Identities

Shifts, periodicity and even/odd formulas

$\sin(\theta + 2\pi) = \sin\theta$	$\cos(\theta + 2\pi) = \cos\theta$	$\tan(\theta + 2\pi) = \tan\theta$
$\sin(\theta + \pi) = -\sin\theta$	$\cos(\theta + \pi) = -\cos\theta$	$\tan(\theta + \pi) = \tan\theta$
$\sin(-\theta) = -\sin\theta$	$\cos(-\theta) = \cos\theta$	$\tan(-\theta) = -\tan\theta$
$\sin\left(\frac{\pi}{2} - \theta\right) = \cos\theta$	$\cos\left(\frac{\pi}{2} - \theta\right) = \sin\theta$	$\tan\left(\frac{\pi}{2} - \theta\right) = \cot\theta$

More Trigonometric Identities

Pythagoras

- $\sin^2\theta + \cos^2\theta = 1$
- $\tan^2\theta + 1 = \sec^2\theta$
- $1 + \cot^2\theta = \csc^2\theta$

Double-angle formulas

- $\sin(2\theta) = 2\sin\theta\cos\theta$
- $\cos(2\theta) = \cos^2\theta - \sin^2\theta = 2\cos^2\theta - 1 = 1 - 2\sin^2\theta$

Half-angle formulas

- $\cos^2\theta = \dfrac{1+\cos(2\theta)}{2}$
- $\sin^2\theta = \dfrac{1-\cos(2\theta)}{2}$

Addition formulas

- $\sin(\theta+\phi) = \sin\theta\cos\phi + \cos\theta\sin\phi$
- $\cos(\theta+\phi) = \cos\theta\cos\phi - \sin\theta\sin\phi$
- $\tan(\theta+\phi) = \dfrac{\tan\theta + \tan\phi}{1 - \tan\theta\tan\phi}$

Example

Find all values of x with $0 \le x \le \pi$ such that $\sin 2x = \sin x$.

Solution:

- Using the double-angle formula $\sin 2x = 2 \sin x \cos x$ we have:

$$2 \sin x \cos x = \sin x$$

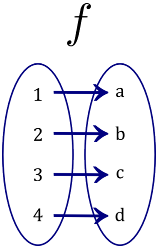
$$2 \sin x \cos x - \sin x = 0$$

$$\sin x (2 \cos x - 1) = 0$$

- Thus, either $\sin x = 0$ or $\cos x = 1/2$.
- For the first case when $\sin x = 0$, we get $x = 0$ or $x = \pi$.
- For the second case when $\cos x = 1/2$, we get $x = \pi/3$.
  - Use the special triangles and CAST rule to get this.
- Thus, we have three solutions: $x = 0$, $x = \pi/3$, $x = \pi$.

Definition of a Function

- A $\boxed{\textbf{function}}$ f, with domain D, is a rule which assigns to each element $x \in D$, a **single** real number which we denote by $f(x)$.
- The $\boxed{\textbf{domain}}$ D is usually a set of real numbers.
- The $\boxed{\textbf{range}}$ of f consists of all possible numbers of the form $f(x)$, with $x \in D$.
- Often we write a function using the symbol $f(x)$, but other common symbols include:
$$y,\ g(x),\ h(x),\ f(t),\ g(t),\ P(x),\ C(x),\ \text{etc.}$$
- You can think of the **domain** of a function as all the **allowed** values of x and the **range** of a function as the **attainable** values of y.
- Below is a diagram of a function f with domain $\{1, 2, 3, 4\}$ and range $\{a, b, c, d\}$.



Descriptions of Functions

- Functions can be described in various ways:
  - **algebraically** (an explicit formula),
  - **visually** (by a graph),
  - **numerically** (by a table of values),
  - **verbally** (by a description in words).
- Using Calculus, you can take an algebraic description of a function and represent it visually.

Example

What does the function

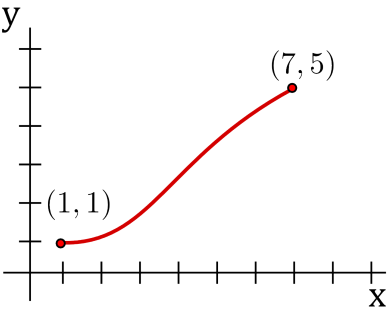
$$f(x) = \frac{2x^2}{x^2 - 1}$$

"look" like in the xy-plane?

Solution: After we learn about derivatives, we will be able to properly sketch this function.

Example

The graph of a function f is shown below:



(a) Estimate $f(4)$.
(b) Determine the domain and range of f.

Solution:

- (a) According to the graph, $f(4)$ is approximately 2.6 ($f(4) \approx 2.6$).

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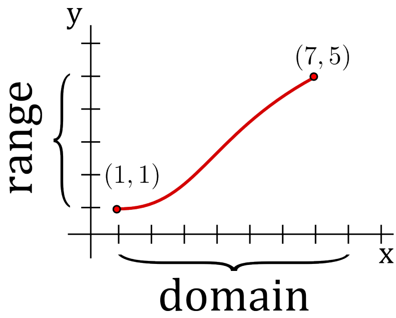
Solution (continued):

- (b) We can determine the domain and range directly from the graph:



- The domain D is the set of all real numbers x between 1 and 7:

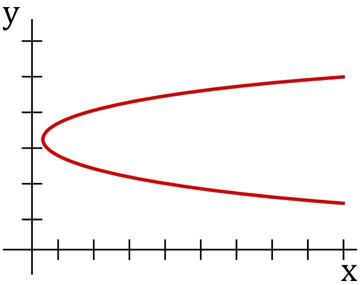
$$D = \{x \in \mathbb{R} \mid 1 \leq x \leq 7\} \quad \text{or} \quad [1, 7].$$

- The range R is the set of all real numbers y between 1 and 5:

$$R = \{y \in \mathbb{R} \mid 1 \leq y \leq 5\} \quad \text{or} \quad [1, 5].$$

Example

Does the following curve represent the graph of a function? Why or why not?



Solution:

- The above is **not** a function since it does not satisfy the definition.
- A function is a rule which assigns a **single** real number to each element in the domain.
- Looking at the above figure, we see that the point $x = 3$ is assigned **two** real numbers.
- Thus, this curve does not represent the graph of a function.

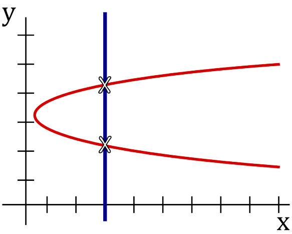
The Vertical Line Test (VLT)

The previous example gives rise to the following test for functions:

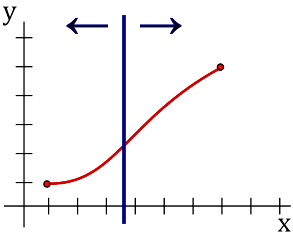
Theorem

The Vertical Line Test

A curve in the xy-plane is the graph of a $\boxed{\textbf{function}}$ of x if and only if **no** vertical line intersects the curve **more than once**.



is not a function



is a function

Finding Domains of Functions

Some **guidelines** to find the domain of a function are:

Domain Restrictions

- You can't **divide by zero**.
  - Be careful with some trigonometric functions that have a hidden division by zero.
  - For example, $\tan x = \dfrac{\sin x}{\cos x}$.
- You can't take the **square root of a negative number**.
  - You also can't take an even root (like a fourth root) of negative numbers.
  - Odd roots (like a cube root) are okay (the cube root of -1 is -1).
- You can't take the **logarithm of a negative number or zero**.
  - We will review logarithms later.
- Other restrictions (e.g., inverse trig functions).

Let's do a **tough** domain problem!

Tough Problem?

Example

Find the domain of the function

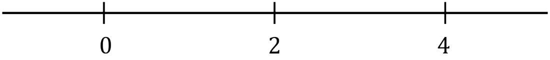
$$f(x) = \frac{|x+2| - \sqrt{x-1}}{\sqrt{x(x-2)(x-4)}}$$

Solution: Let us split the problem up into four subproblems:

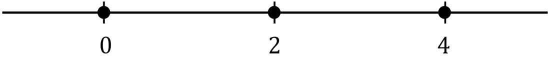
1. The $|x+2|$ has no effect on the domain as any value of x is allowed when dealing with absolute values.
2. The $\sqrt{x-1}$ implies that we require $x - 1 \geq 0$ as we cannot take the square root of a negative number.
3. Similarly, we require $x(x-2)(x-4) \geq 0$ since $x(x-2)(x-4)$ appears under a square root.
4. Note that the denominator is zero when $x = 0$, $x = 2$ or $x = 4$. Therefore, these three numbers are **NOT** in the domain.

Solve $x(x-2)(x-4) \geq 0$

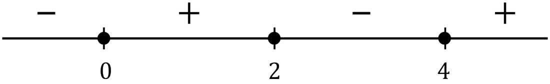
The split points are 0, 2 and 4:



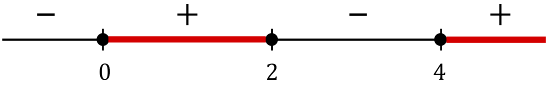
Note that each of $0, 2, 4$ give $0 \geq 0$, so we put closed circles:



Now choose test points. At $x = -1$ it is negative. At $x = 1$ it is positive. At $x = 3$ it is negative. At $x = 5$ it is positive.



Since we need $x(x-2)(x-4) \geq 0$, we look at where the $+$ signs are:



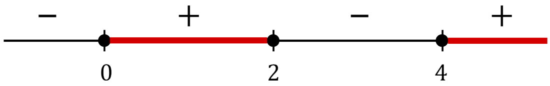
Therefore, the solution to $x(x-2)(x-4) \geq 0$ is $[0, 2] \cup [4, \infty)$.

Solution

Summarizing, #1, #2, #3 and #4 give the following conditions on x:

1. No condition
2. $x \geq 1$.
3. x is in $[0, 2] \cup [4, \infty)$.
4. x cannot equal 0, 2 or 4.

Combining these together give the domain to be $[1, 2) \cup (4, \infty)$.
To see this, look at the last number line:



Remove the points 0, 2 and 4, also remove everything that is less than 1 but keep anything ≥ 1 (by bullet point #2), then what remains is precisely $[1, 2) \cup (4, \infty)$.

Examples of Functions

There are many types of functions that we deal with in this course:

- polynomial functions,
- the absolute value function,
- the square root function,
- exponential functions,
- trigonometric functions,
- the logarithmic function, etc.

Some functions we have already seen!

Linear functions, vertical parabolas and trigonometric functions (note that some conics, such as a circle or hyperbola, are **not** functions as they don't satisfy the vertical line test!).

Polynomial Functions

- Any function of the form:

$$f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$$

  where n is a nonnegative integer and the a_i's are fixed real numbers is called a **polynomial**.
- The constants a_i are called the **coefficients** of the polynomial.
- If the leading coefficient $a_n \neq 0$, then we say the polynomial has **degree** n.
- Any real number is in the domain of a polynomial, that is, the domain D of a polynomial is

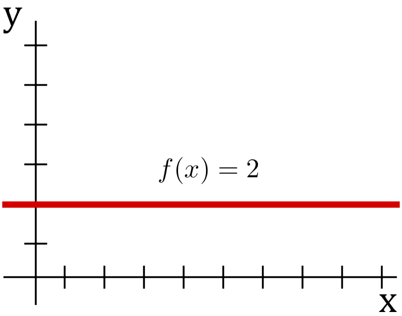
$$D = \{x \in \mathbb{R}\}.$$

- The range depends on the polynomial and can be determined by using calculus.

The following slides demonstrate some special cases of polynomials...

Constant Functions

- A $\boxed{\textbf{constant function}}$ is any function of the form $f(x) = a$ where $a \in \mathbb{R}$.
- Note that a constant function is a polynomial of degree 0.
- A visual representation of a constant function is a horizontal line in the xy-plane.
- For example, $f(x) = 2$ is a constant function and is plotted below.



Linear and Quadratic Functions

Linear functions:

- A **linear function** is any function of the form

$$f(x) = mx + b$$

  where $m, b \in \mathbb{R}$ are any fixed real numbers.
- We already reviewed linear functions (**straight lines**) in a previous lecture.
- A linear function is a polynomial of degree 1.

Quadratic Functions:

- A $\boxed{\textbf{quadratic function}}$ is any function of the form:

$$f(x) = ax^2 + bx + c$$

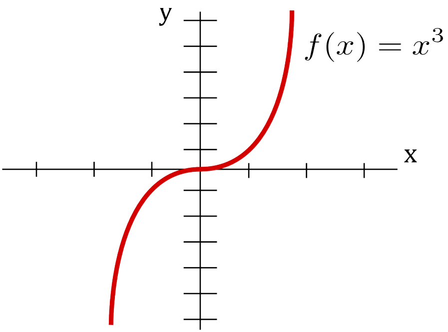
  where $a, b, c \in \mathbb{R}$ are any fixed real numbers and $a \neq 0$.
- These functions are simply **vertical parabolas**, which we reviewed in a previous lecture.
- A quadratic function is a polynomial of degree 2.

Cubic Functions

- A $\boxed{\textbf{cubic function}}$ is any function of the form:

$$f(x) = ax^3 + bx^2 + cx + d$$

  where $a, b, c, d \in \mathbb{R}$ are any fixed real numbers and $a \neq 0$.
- For example, $f(x) = x^3 - x + 1$ is a cubic function.
- A cubic function is a polynomial of degree 3.
- The function $y = x^3$ is sketched below:

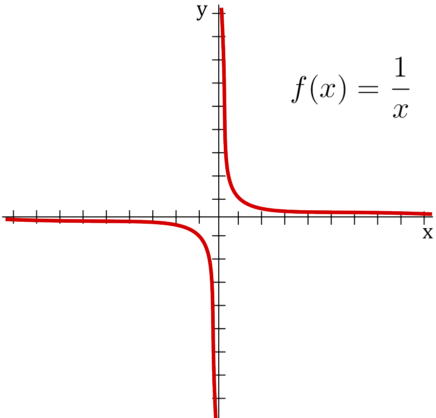


Rational Functions

- A **rational function** is is a ratio of two polynomials:

$$f(x) = \frac{p(x)}{q(x)}.$$

- Here p and q are polynomials.
- The domain is all x such that $q(x) \neq 0$.
- When $p(x) = 1$ and $q(x) = x$ we get the rational function $f(x) = \dfrac{1}{x}$:



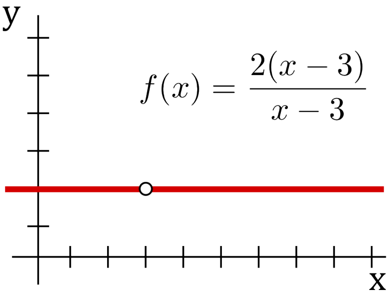
Example

Sketch the rational function

$$f(x) = \frac{2(x-3)}{x-3}.$$

Solution:

- Note that $x = 3$ gives a division by zero, thus $f(x)$ has domain $\{x \in \mathbb{R} \mid x \neq 3\}$.
- If you cancel the $x - 3$ terms you end up with $f(x) = 2$.
- Thus, for $x \neq 3$ the function looks like the constant function $f(x) = 2$.
- At $x = 3$ we have a hole (the point $(x, y) = (3, 2)$ is deleted from the graph $f(x) = 2$).



Piecewise-Defined Functions

- A **piecewise-defined function** is defined by different formulas in parts of its domain.
- That is, it is defined in separate **pieces**.

Example

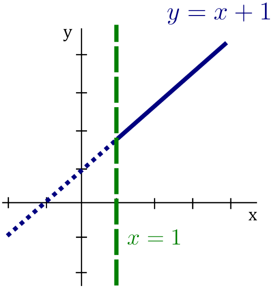
Sketch the following piecewise defined function:

$$f(x) = \begin{cases} x, & \text{if } x \leq 1, \\ x+1, & \text{if } x > 1. \end{cases}$$

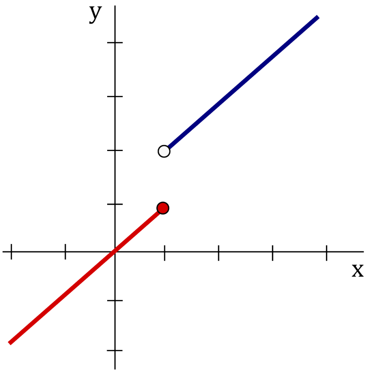
Solution: To get a visual representation of f we observe that

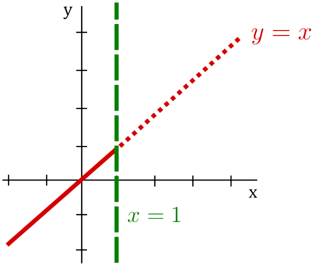
- If $x \leq 1$, then $f(x) = x$, so the part of the graph of f that lies to the left of the vertical line $x = 1$ must coincide with the line $y = x$.
- If $x > 1$, then $f(x) = x + 1$, so the part of the graph of f that lies to the right of the line $x = 1$ must coincide with $y = x + 1$.

Sketch on the next slide...



$$f(x) = \begin{cases} x & \text{if } x \leq 1 \\ x + 1 & \text{if } x > 1 \end{cases}$$





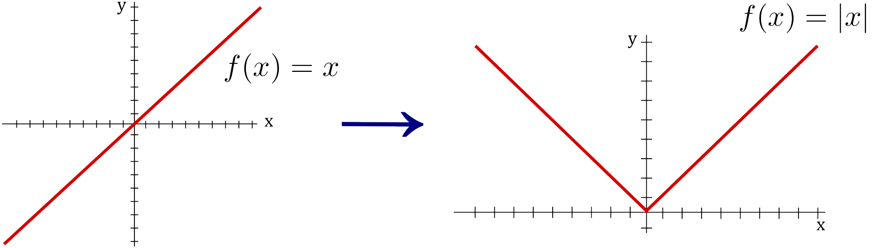
The Absolute Value Function

Definition

The **absolute value function** is defined by:

$$|x| = \begin{cases} x, & \text{if } x \geq 0, \\ -,x & \text{if } x < 0. \end{cases}$$

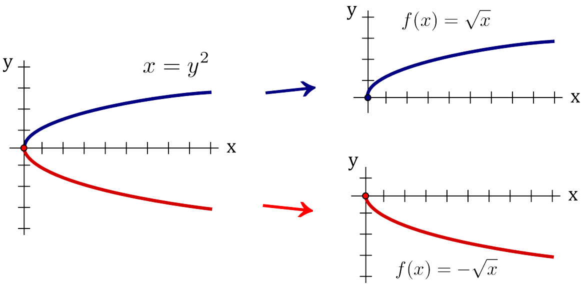
To visualize, take the straight line $f(x) = x$ and **“reflect”** the negative part to make it positive:



Root Functions and Power Functions

Root functions:

- The function $f(x) = x^{1/n} = \sqrt[n]{x}$ is called a $\boxed{\textbf{root function}}$, here n is a positive integer.
- For $n = 2$, it is the $\boxed{\textbf{square root function}}$ $f(x) = \sqrt{x}$.
- $\sqrt{x}$ means the **positive** square root of x.

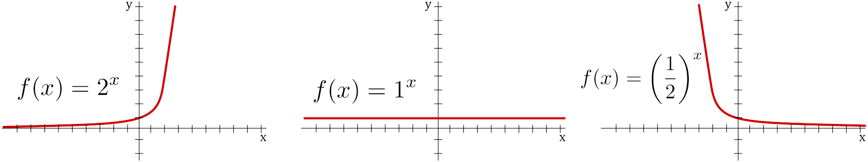


Power Functions:

- If a is any fixed **real number**, then $f(x) = x^a$ is called $\boxed{\textbf{power function}}$.
- In the special case that a is a **positive integer**, then $f(x) = x^a$ is a **polynomial** with leading coefficient equal to 1 and all other coefficients equal to 0.
- If $a = -1$, then $f(x) = x^{-1} = 1/x$ is a **rational function** as described previously.
- If a is a fraction $1/n$ where n is a positive integer, then we get a **root function**.

Exponential Functions

A function of the form $f(x) = a^x$, where the base a is a fixed positive real number is called an **exponential function**. It can be visually represented as:



More on this later!

Even and Odd Functions

- A function f is an **even function** if $f(-x) = f(x)$, for all x in the domain of f.
- A function f is an **odd function** if $f(-x) = -f(x)$, for all x in the domain of f.

Example

The functions $f(x) = |x|$, $f(x) = \cos x$, and $f(x) = x^2$ are all **even**.

Graphical Interpretation of Even Functions

If you **reflect** the graph about the y-axis, the graph stays the same!

Example

The functions $f(x) = 1/x$, $f(x) = \sin x$, and $f(x) = x^3$ are all **odd**.

Graphical Interpretation of Odd Functions

If you **rotate** the graph 180° about the origin, the graph stays the same!

Example

Determine whether the function $f(x) = 2x - x^2$ is even, odd, or neither even nor odd.

Solution:

- We have

$$\begin{aligned} f(-x) &= 2(-x) - (-x)^2 \\ &= -2x - x^2 \end{aligned}$$

- Note that $f(-x) \neq f(x)$, so the function is not even.
- Additionally,

$$\begin{aligned} -f(x) &= -(2x - x^2) \\ &= -2x + x^2. \end{aligned}$$

- Thus, $f(-x) \neq -f(x)$, so the function is not odd.
- Therefore, $f(x)$ is **neither** even nor odd.

Increasing and Decreasing Functions

Definition

A function f is called $\boxed{\text{increasing}}$ on an interval I if

$$f(x_1) < f(x_2) \qquad \text{whenever } x_1 < x_2 \text{ in } I.$$

A function f is called $\boxed{\text{decreasing}}$ on an interval I if

$$f(x_1) > f(x_2) \qquad \text{whenever } x_1 < x_2 \text{ in } I.$$

- Note that to be increasing the inequality $f(x_1) < f(x_2)$ must be satisfied for **every pair** of numbers x_1, x_2 in the interval I with $x_1 < x_2$.
- Visually, as you go from left to right, increasing on an interval means **"going up"** and decreasing on an interval means **"going down"**.

Example

The vertical parabola $f(x) = x^2$ is increasing on $[0, \infty)$ and decreasing on $(-\infty, 0]$.

Transformations

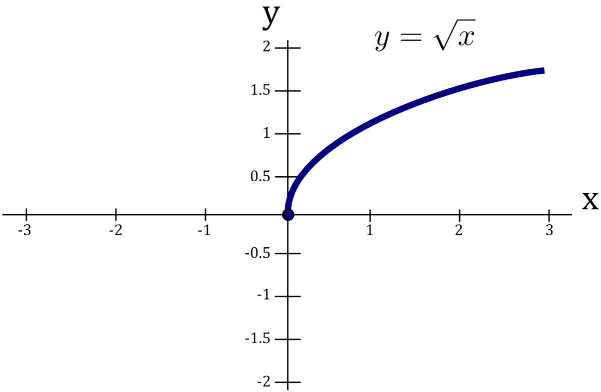
- One way to create a new function from a known function is to apply a sequence of transformations.
- Types of transformations include:
  - Translations
  - Stretches
  - Reflections

Function	condition	How to graph $F(x)$ given the graph of $f(x)$
$F(x) = f(x) + c$	$c > 0$	Shift $f(x)$ upwards by c units
$F(x) = f(x) - c$	$c > 0$	Shift $f(x)$ downwards by c units
$F(x) = f(x + c)$	$c > 0$	Shift $f(x)$ to the left by c units
$F(x) = f(x - c)$	$c > 0$	Shift $f(x)$ to the right by c units
$F(x) = af(x)$	$a > 0$	Stretch $f(x)$ vertically by a factor of a
$F(x) = f(bx)$	$b > 0$	Stretch $f(x)$ horizontally by a factor of $1/b$
$F(x) = -f(x)$		Reflect $f(x)$ about the x-axis
$F(x) = f(-x)$		Reflect $f(x)$ about the y-axis
$F(x) = \|f(x)\|$		Take the part of the graph of $f(x)$ that lies below the x-axis and reflect it about the x-axis

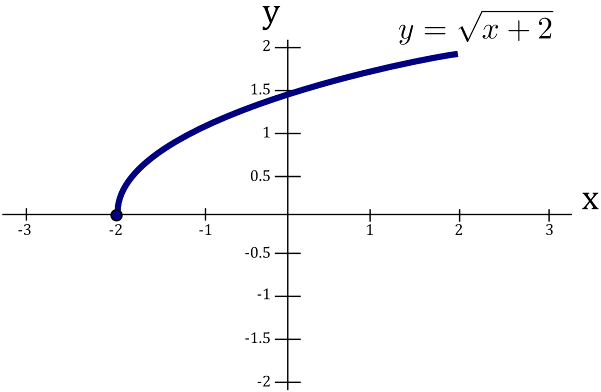
Example

Use appropriate transformations to sketch the graph of the function $y = |\sqrt{x+2} - 1| - 1$.

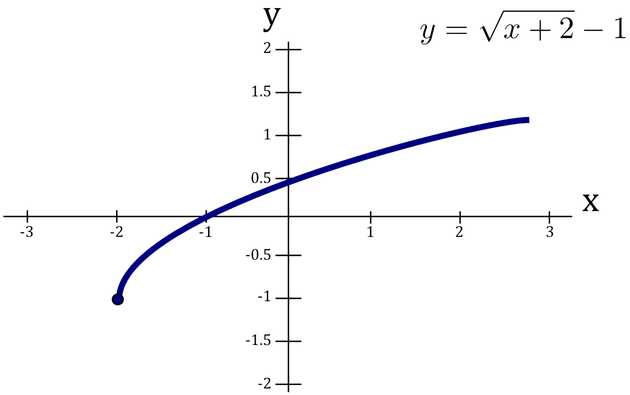
Solution: We start with the graph of a function we know how to sketch, in particular, $y = \sqrt{x}$:



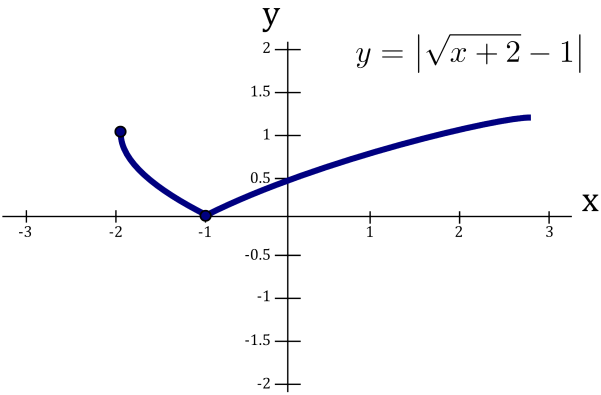
To obtain the graph of the function $y = \sqrt{x+2}$ from the graph $y = \sqrt{x}$, we must shift $y = \sqrt{x}$ to the left by 2 units:



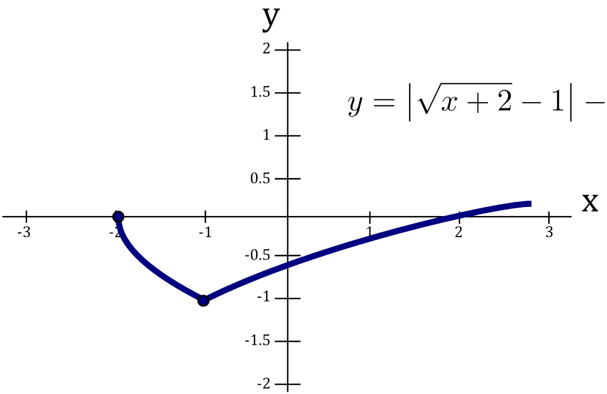
To obtain the graph of the function $y = \sqrt{x+2} - 1$ from the graph $y = \sqrt{x+2}$, we must shift $y = \sqrt{x+2}$ downwards by 1 unit:



To obtain the graph of the function $y = |\sqrt{x+2} - 1|$ from the graph $y = \sqrt{x+2} - 1$, we must take the part of the graph of $y = \sqrt{x+2} - 1$ that lies below the x-axis and reflect it (upwards) about the x-axis:



To obtain the graph of the function $y = |\sqrt{x+2} - 1| - 1$ from the graph $y = |\sqrt{x+2} - 1|$, we must shift $y = |\sqrt{x+2} - 1|$ downwards by 1 unit:



We now have the graph of the function $y = |\sqrt{x+2} - 1| - 1$.

Combining Functions

- Let f and g be two functions.
- Then we can form new functions by adding, subtracting, multiplying or dividing.
- These new functions, $f+g$, $f-g$, fg and f/g, are defined in the obvious manner:

$$(f+g)(x) = f(x)+g(x) \qquad (f-g)(x) = f(x)-g(x)$$

$$(fg)(x) = f(x)g(x) \qquad \left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$$

- Suppose D_f is the domain of f and D_g is the domain of g.
- Then the domains of $f+g$, $f-g$ and fg are the same and are equal to the intersection:

$$D_f \cap D_g.$$

  That is, everything that is in **common** to both the domain of f **and** the domain of g.
- Since division by zero is **not allowed**, the domain of f/g is

$$\{x \in D_f \cap D_g : g(x) \neq 0\}.$$

  That is, everything that is in **common** to both the domain of f **and** the domain of g. except the points that make $g(x)$ equal to zero.

Composition

- Given two functions f and g, the $\boxed{\textbf{composition}}$ of f and g, denoted by $f \circ g$, is defined as:

$$(f \circ g)(x) = f(g(x)).$$

- The domain of $f \circ g$ is

$$\{x \in D_g : g(x) \in D_f\}.$$

  That is, it contains all values x in the domain of g such that $g(x)$ is in the domain of f.

Example

Let $f(x) = x^2$ and $g(x) = \sqrt{x}$. Find the domain of $f \circ g$.

Solution:

- The domain of f is $D_f = \{x \in \mathbb{R}\}$.
- The domain of g is $D_g = \{x \in \mathbb{R} : x \geq 0\}$.
- The function $(f \circ g)(x) = f(g(x))$ is:

$$f(g(x)) = \left(\sqrt{x}\right)^2 = x.$$

- Typically, $h(x) = x$ would have a domain of $\{x \in \mathbb{R}\}$, but since it came from a **composition**, we must consider $g(x)$ when looking at the domain of $f(g(x))$.
- Thus, the domain of $f \circ g$ is $\{x \in \mathbb{R} : x \geq 0\}$.

Example

Let $f(x) = x^2 + 3$ and $g(x) = x - 2$. Find $f + g$, $f - g$, fg, and f/g.

Solution: We have

$$\begin{aligned}(f+g)(x) &= f(x) + g(x) \\ &= (x^2+3) + (x-2) \\ &= x^2 + x + 1\end{aligned}$$

$$\begin{aligned}(f-g)(x) &= f(x) - g(x) \\ &= (x^2+3) - (x-2) \\ &= x^2 + 3 - x + 2 \\ &= x^2 - x + 5\end{aligned}$$

$$\begin{aligned}(fg)(x) &= f(x) \cdot g(x) \\ &= (x^2+3)(x-2) \\ &= x^3 - 2x^2 + 3x - 6\end{aligned}$$

$$\begin{aligned}\left(\frac{f}{g}\right)(x) &= \frac{f(x)}{g(x)} \\ &= \frac{x^2+3}{x-2}\end{aligned}$$

Example

Let $f(x) = x^2 + 3$ and $g(x) = x - 2$.
Find $f \circ g$ and $g \circ f$.

Solution: We have

$$\begin{aligned}
(f \circ g)(x) &= f(g(x)) \\
&= f(x-2) \\
&= (x-2)^2 + 3 \\
&= x^2 - 4x + 7
\end{aligned}$$

$$\begin{aligned}
(g \circ f)(x) &= g(f(x)) \\
&= g(x^2 + 3) \\
&= (x^2 + 3) - 2 \\
&= x^2 + 1
\end{aligned}$$

Observation: In general (i.e., for most functions f and g), $f(g(x)) \neq g(f(x))$.

Exponential Functions

An $\boxed{\textbf{exponential function}}$ is a function of the form $f(x) = a^x$.

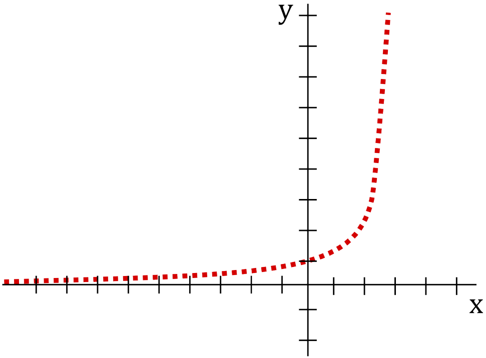
Here a is a positive real number and x is the input.

To define the exponential function we look at various kinds of input values.

- **Zero Exponent:** $a^0 = 1$.
- **Whole Number Exponent:** $a^n = \underbrace{a \cdot a \cdots a}_{n \text{ times}}$, where $n = 1, 2, 3, \ldots$
- **Integer Exponent:** $a^{-n} = \dfrac{1}{a^n}$, where $n = 1, 2, 3, \ldots$
- **Rational Exponent:** $a^{\frac{p}{q}} = \sqrt[q]{a^p}$, where $q = 1, 2, 3, \ldots$, and $p = 0, \pm 1, \pm 2, \pm 3, \ldots$.

Irrational Exponents?

- If x is an **irrational** number, how do you define a^x?
- What is the meaning of $3^{\sqrt{2}}$ or 2^{π}?
- Looking at only rational values of x, the function $y = a^x$ is "sketched" below (for $a > 1$).
- As you can see, there are gaps in the graph at the irrationals.



- To define a^x for irrational x, we use a limiting process to fill in the gaps.
- Consider the expression $3^{\sqrt{2}}$.
- Note that

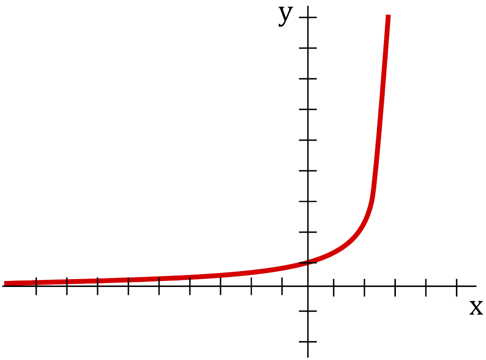
$$\sqrt{2} \approx 1.4142135623730950488016887242096980785696718753769\ldots$$

- We now approximate $3^{\sqrt{2}}$ by taking approximations of $\sqrt{2}$:

$$\begin{aligned}
3^{1.4} &= \underline{\mathbf{4.}}655536722\ldots \\
3^{1.41} &= \underline{\mathbf{4.7}}06965002\ldots \\
3^{1.414} &= \underline{\mathbf{4.72}}7695035\ldots \\
3^{1.4142} &= \underline{\mathbf{4.728}}733930\ldots \\
3^{1.41421} &= \underline{\mathbf{4.728}}785881\ldots \\
3^{1.414213} &= \underline{\mathbf{4.72880}}1466\ldots \\
3^{1.4142135} &= \underline{\mathbf{4.728804}}064\ldots \\
&\vdots \\
3^{\sqrt{2}} &= \underline{\mathbf{4.728804388}}\ldots
\end{aligned}$$

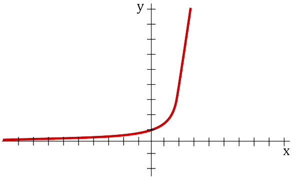
Using better and better approximations for $\sqrt{2}$ we can define the expression $3^{\sqrt{2}}$ as the resulting number in this limiting process.

After doing this for all the irrational numbers, this essentially this fills in the gaps in the graph above to give the following:



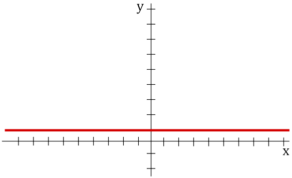
There are **three kinds** of exponential functions $f(x) = a^x$ depending on whether:

- $a > 1$,
- $a = 1$, or
- $0 < a < 1$.

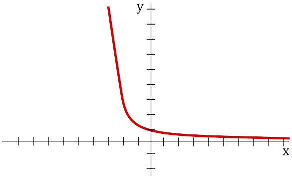


$f(x) = a^x$

$a > 1$



$f(x) = 1^x$



$f(x) = a^x$

$0 < a < 1$

Properties of Exponential Functions

- **Only defined for positive a:** a^x is only defined for all real x if $a > 0$
- **Always positive:** $a^x > 0$, for all x
- **Exponent rules:**
  - $a^x a^y = a^{x+y}$
  - $\dfrac{a^x}{a^y} = a^{x-y}$
  - $(a^x)^y = a^{xy}$
  - $a^x b^x = (ab)^x$
- **Long term behaviour:** If $a > 1$, then $a^x \to \infty$ as $x \to \infty$ and $a^x \to 0$ as $x \to -\infty$.

The last property can be observed from the graph.

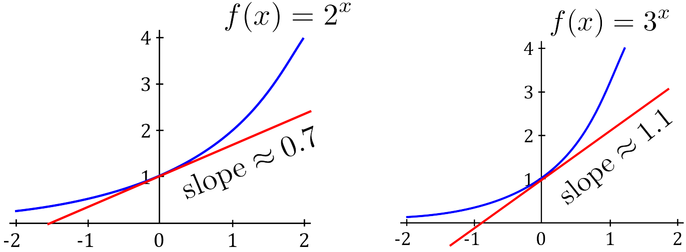
A Special Base

- Of all the bases that we can choose there is one that is the most convenient.
- Consider the function $f(x) = a^x$ and a tangent line at the point $(0, 1)$.

Question

What base has the property that at the point $(0, 1)$ the slope of the tangent line is one?

- If $a = 2$, the slope of the tangent line is approximately 0.7.
- If $a = 3$ the slope of the tangent line is approximately 1.1.

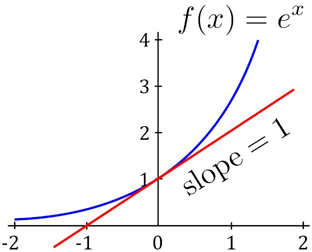


A Special Base

- It turns out that when the base is approximately

$$2.71828182845904523536028747135266249775724709369995\ldots$$

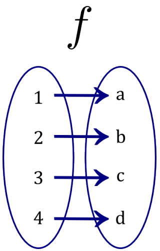
the slope of the tangent line is exactly **equal** to one!

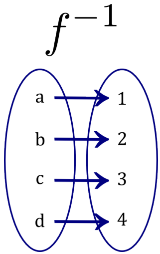


- This number is denoted by $e = 2.71828\ldots$ and is an irrational number.
- It is sometimes called **Euler's constant** named after the mathematician Leonhard Euler.
- Its **importance** stems from the fact that it simplifies many formulas of Calculus and also shows up in other fields of mathematics.

Inverse Functions

- An $\boxed{\textbf{inverse}}$ (function) is a function that **undoes** another function.
- That is, if $f(x)$ produces y, then putting y into the inverse gives x.
- A function f that has an inverse is called $\boxed{\textbf{invertible}}$.
- We denote the inverse of f by f^{-1}.

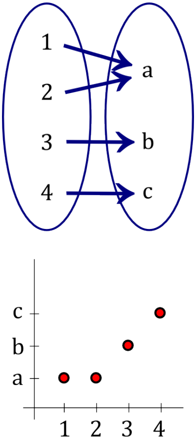




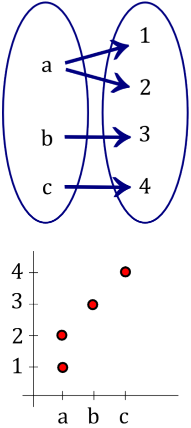
- Notice how f maps 1 to a, and f^{-1} undoes this.
- **Caution:** Don't confuse $f^{-1}(x)$ with reciprocals: the inverse f^{-1} is **different** from $\frac{1}{f(x)}$.

Question

Does **every** function have an inverse function?



is a function



is not a function

- The above diagrams show that **not** every function has an inverse function.
- The diagram on the left is a function.
- However, the diagram on the right does the **opposite** and is not a function.
- An inverse needs to be a function, thus, the left diagram is a function that is not invertible.

When does f have an inverse?

It is easy to see that if a function $f(x)$ is going to have an inverse, then:

$f(x)$ **never** takes on the same value twice.

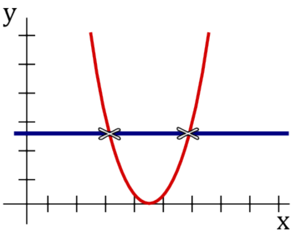
A function $f(x)$ is called **one-to-one** if it never takes on the same value twice, that is,

$$f(x_1) \neq f(x_2) \quad \text{whenever } x_1 \neq x_2.$$

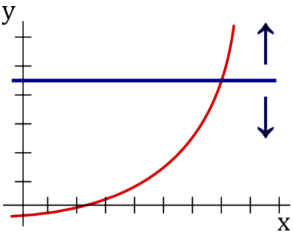
Theorem

Horizontal Line Test

A function is **one-to-one** if and only if no horizontal line intersects its graph more than once.



is not one-to-one



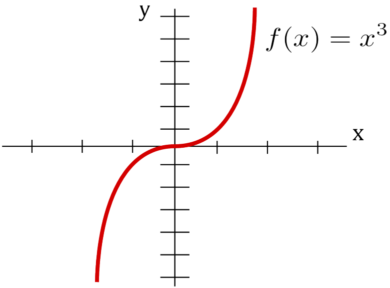
is one-to-one

Example

The parabola $f(x) = x^2$ it **not** one-to-one because it does not satisfy the horizontal line test. For example, the horizontal line $y = 1$ intersects the parabola at two points ($x = \pm 1$).

Example

The function $f(x) = x^3$ is **one-to-one** since it satisifies the horizontal line test.



Formal Definition for an Inverse Function

Let f be a **one-to-one** function with domain A and range B.

Its $\boxed{\textbf{inverse function } f^{-1}}$ has domain B and range A and is defined by

$$f^{-1}(y) = x \iff f(x) = y, \quad \text{for any } y \text{ in } B.$$

That is, if f maps x to y, then f^{-1} maps y back to x.

This gives rise to the **cancellation formulas**:

$$f^{-1}(f(x)) = x, \quad \text{for every } x \text{ in } A,$$

$$f(f^{-1}(x)) = x, \quad \text{for every } x \text{ in } B.$$

Example

If $f(x) = x^9 + 2x^7 + x + 1$, find $f^{-1}(-3)$, $f^{-1}(5)$ and $f^{-1}(1)$.

Solution:

- Rather than trying to compute a formula for f^{-1} we make a table of values for $f(x)$:

x	-2	-1	0	1	2
$f(x)$	-769	-3	1	5	771

- Going down the table gives f.
- Going up the table (**the opposite**) gives f^{-1}.
- That is, $f^{-1}(-3) = -1$, $f^{-1}(5) = 1$ and $f^{-1}(1) = 0$.
- This is because $f(a) = b$ is equivalent to $a = f^{-1}(b)$).
- Note that if we did not get an output of $f(x)$ to be -3, 5, or 1, we would need to make a larger table of values.
- Otherwise, this technique of finding the inverse at specific points will not work.

Equation for Inverse: To compute the inverse of a function we use the following **guidelines**:

1. Write down $y = f(x)$.
2. Solve for x in terms of y.
3. Switch the x's and y's.
4. The result is $y = f^{-1}(x)$.

Example

We find the inverse of the function $f(x) = 2x^3 + 1$.

1. Start with $y = f(x)$: $\quad y = 2x^3 + 1$

2. Solve for x: $\quad y - 1 = 2x^3$

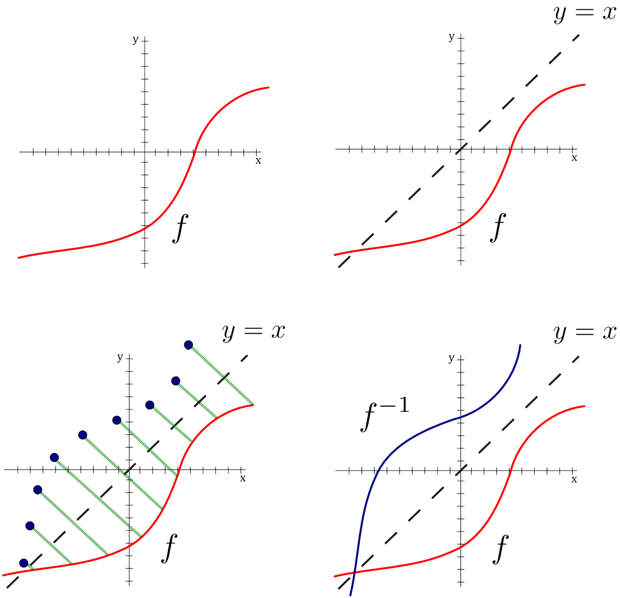
$$\frac{y-1}{2} = x^3$$

$$x = \sqrt[3]{\frac{y-1}{2}}$$

3. Switch x and y: $\quad y = \sqrt[3]{\frac{x-1}{2}}$

4. Write $f^{-1}(x)$: $\quad f^{-1}(x) = \sqrt[3]{\frac{x-1}{2}}$

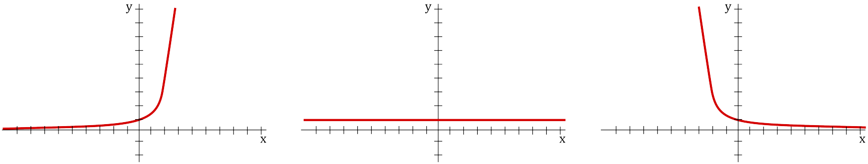
Graphing Inverses: To obtain $f^{-1}(x)$ **reflect** about $y = x$.



Logarithms

Recall the **three kinds** of exponential functions $f(x) = a^x$ depending on whether:

- $a > 1$,
- $a = 1$, or
- $0 < a < 1$.



$f(x) = a^x$ $\quad$ $a > 1$ $\qquad$ $f(x) = 1^x$ $\qquad$ $f(x) = a^x$ $\quad$ $0 < a < 1$

- So long as $a \neq 1$, the function $f(x) = a^x$ satisfies the horizontal line test.
- Thus, for $a \neq 1$, $f(x) = a^x$ has an inverse function.
- We call the **inverse of a^x** the **logarithmic function with base a** and denote it by $\log_a$:

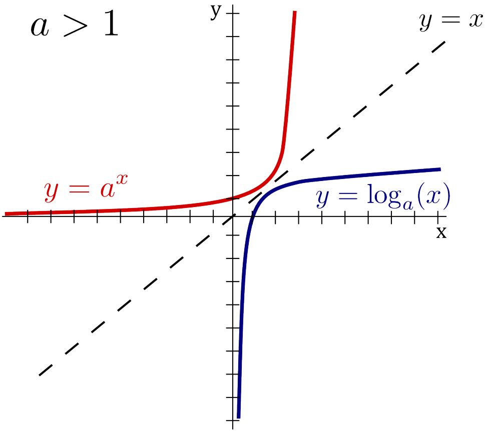
$$\log_a x = y \iff a^y = x.$$

- The **cancellation formulas** for logs are:

$$\log_a(a^x) = x, \quad \text{for every } x \in \mathbb{R},$$

$$a^{\log_a(x)} = x, \quad \text{for every } x > 0.$$

- The function $f(x) = a^x$ for $a \neq 1$ has domain $\mathbb{R}$ and range $(0, \infty)$.
- Thus, the logarithmic function has domain $(0, \infty)$ and range $\mathbb{R}$.



Log Properties

Theorem

Log Properties

Let A, B be **positive numbers** and $a > 0$ $(a \neq 1)$ be a base. Then

- $\log_a(AB) = \log_a A + \log_a B$,
- $\log_a\left(\dfrac{A}{B}\right) = \log_a A - \log_a B$,
- $\log_a(A^n) = n\log_a A$, where n is any real number.

Example

Compute $\log_2(24) - \log_2(3)$.

Solution: To compute $\log_2(24) - \log_2(3)$ we can do the following:

$$\log_2(24) - \log_2(3) = \log_2\left(\frac{24}{3}\right) = \log_2(8) = 3,$$

since $2^3 = 8$.

The natural logarithm

- Since the number $e \approx 2.71828$ is the most convenient base to use in Calculus, we give the logarithm with base e a special name:

  the natural logarithm.

- We also give it special notation:
$$\log_e x = \ln x.$$

- **How do you pronounce ln?**
  - You may pronounce it as either: "el - en", "lawn" or "natural log".

- The log properties also hold for $\ln x$: Let A, B be **positive numbers**.
  - $\ln(AB) = \ln A + \ln B$,
  - $\ln\left(\dfrac{A}{B}\right) = \ln A - \ln B$,
  - $\ln(A^n) = n \ln A$, where n is any real number.

- Often we need to turn a logarithm (in base a) into a natural logarithm.

- This gives rise to the **change of base formula:**
$$\log_a x = \frac{\ln x}{\ln a}.$$

Example

Write $\ln A + 2\ln B - \ln C$ as a single logarithm.

Solution: Using properties of logarithms, we have,

$$\begin{aligned}
\ln A + 2\ln B - \ln C &= \ln A + \ln B^2 - \ln C \\
&= \ln(AB^2) - \ln C \\
&= \ln \frac{AB^2}{C}.
\end{aligned}$$

Example

If $e^{x+2} = 6e^{2x}$, then solve for x.

Solution:
Since both sides are positive, we can take the natural logarithm of both sides.

By the cancellation formulas (along with $\ln e = 1$), we have:

$$\begin{aligned} e^{x+2} &= 6e^{2x} \\ \ln e^{x+2} &= \ln(6e^{2x}) \\ x+2 &= \ln 6 + \ln e^{2x} \\ x+2 &= \ln 6 + 2x \\ x &= 2 - \ln 6. \end{aligned}$$

Example

If $\ln(2x-1) = 2\ln(x)$, then solve for x.

Solution:
Taking "e" of both sides and noting the cancellation formulas, we have:

$$\begin{aligned} e^{\ln(2x-1)} &= e^{2\ln(x)} \\ (2x-1) &= e^{\ln(x^2)} \\ 2x-1 &= x^2 \\ x^2-2x+1 &= 0 \\ (x-1)^2 &= 0 \end{aligned}$$

Therefore, the solution is $x = 1$.

Example

Find the domain of $f(x) = \dfrac{1}{\sqrt{e^x + 1}}$.

Solution:

- For domain, we cannot divide by zero or take the square root of negative numbers.
- But exponentials are always positive!
- Thus, $e^x + 1 > 0$.
- Therefore, $e^x + 1$ is never zero nor negative, and gives no restrictions on x.
- Thus, the domain is $\mathbb{R}$.

Example

Find the exponential function $f(x) = k\,a^x$ that passes through the points $(1, 6)$ and $(2, 18)$.

Solution:

- We substitute our two points into the equation to get:

$$x = 1, y = 6 \quad \Longrightarrow \quad 6 = ka^1$$

$$x = 2, y = 18 \quad \Longrightarrow \quad 18 = ka^2$$

- This gives us $6 = ka$ and $18 = ka^2$.
- The first equation is $k = 6/a$ and subbing this into the second gives:

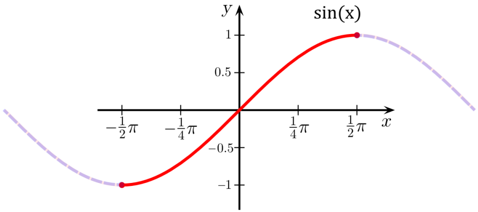
$$18 = \frac{6}{a}a^2.$$

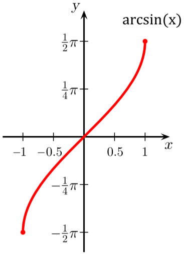
- Thus, $18 = 6a$ and $a = 3$.
- Now we can see from $6 = ka$ that $k = 2$.
- Therefore, the exponential function is

$$f(x) = 2 \cdot 3^x.$$

Inverse Sine Function

- **Issue:** The function $y = \sin x$ is **NOT** one-to-one, and hence, not invertible.
- It'd be nice to have the concept of an inverse sine function.
- To overcome this difficulty, we restrict the domain so it becomes one-to-one.
- There are many intervals that work. The **convention** is to use $[-\pi/2,\ \pi/2]$.
- The graph of $y = \sin x$ restricted to $[-\pi/2,\ \pi/2]$ is one-to-one:

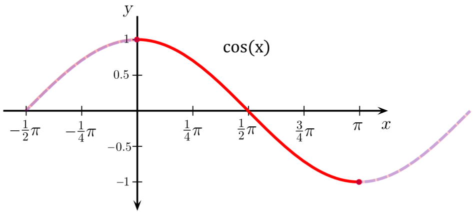


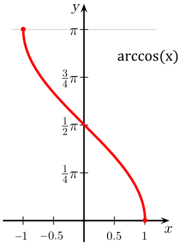


- **Definition:** The $\boxed{\textbf{inverse sine function}}$ (or $\boxed{\textbf{arcsin}}$) is defined to be the inverse of the restricted sine function (that is, $y = \sin x$ restricted to $[-\pi/2,\ \pi/2]$).
- **Domain** of $\sin^{-1}(x)$: $[-1,\ 1]$ **Range** of $\sin^{-1}(x)$: $[-\pi/2,\ \pi/2]$

Inverse Cosine Function

- **Issue:** The function $y = \cos x$ is **NOT** one-to-one, and hence, not invertible.
- It'd be nice to have the concept of an inverse cosine function.
- To overcome this difficulty, we restrict the domain so it becomes one-to-one.
- There are many intervals that work. The **convention** is to use $[0,\ \pi]$.
- The graph of $y = \cos x$ restricted to $[0,\ \pi]$ is one-to-one:

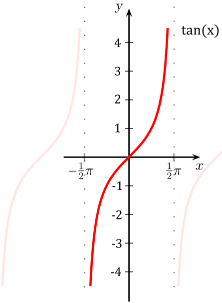


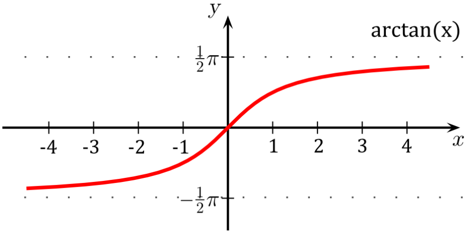


- **Definition:** The $\boxed{\textbf{inverse cosine function}}$ (or $\boxed{\textbf{arccos}}$) is defined to be the inverse of the restricted cosine function (that is, $y = \cos x$ restricted to $[0,\ \pi]$).
- **Domain** of $\cos^{-1}(x)$: $[-1,\ 1]$ **Range** of $\cos^{-1}(x)$: $[0,\ \pi]$

Inverse Tangent Function

- **Issue:** The function $y = \tan x$ is **NOT** one-to-one, and hence, not invertible.
- It'd be nice to have the concept of an inverse tangent function.
- To overcome this difficulty, we restrict the domain so it becomes one-to-one.
- There are many intervals that work. The **convention** is to use $\left(-\dfrac{\pi}{2},\ \dfrac{\pi}{2}\right)$.
- The graph of $y = \tan x$ restricted to $\left(-\dfrac{\pi}{2},\ \dfrac{\pi}{2}\right)$ is one-to-one:





- **Definition:** The $\boxed{\textbf{inverse tangent function}}$ (or $\boxed{\textbf{arctan}}$) is defined to be the inverse of the restricted tangent function (that is, $y = \tan x$ restricted to $\left(-\frac{\pi}{2},\ \frac{\pi}{2}\right)$).
- **Domain** of $\tan^{-1}(x)$: $(-\infty,\ \infty)$ **Range** of $\tan^{-1}(x)$: $\left(-\dfrac{\pi}{2},\ \dfrac{\pi}{2}\right)$

Cancellation Rules

Since we restricted the domains of the trigonometric functions before defining **inverse trig** functions, the cancellation rules must take this into account.

Theorem

Cancellation Rules for Inverse Trig

$$\sin(\sin^{-1} x) = x, \quad x \in [-1, 1] \qquad \sin^{-1}(\sin x) = x, \quad x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

$$\cos(\cos^{-1} x) = x, \quad x \in [-1, 1] \qquad \cos^{-1}(\cos x) = x, \quad x \in [0, \pi]$$

$$\tan(\tan^{-1} x) = x, \quad x \in (-\infty, \infty) \qquad \tan^{-1}(\tan x) = x, \quad x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

Arcsine Example

Example

Find $\sin^{-1}(0)$, $\sin^{-1}(1)$ and $\sin^{-1}(-1)$.

Solution:

- These simply comes from the graph:

$$\sin^{-1}(0) = 0 \qquad \sin^{-1}(1) = \frac{\pi}{2} \qquad \sin^{-1}(-1) = -\frac{\pi}{2}$$

Example

Find $\sin^{-1}(1/2)$.

Solution:

- **Range**: As $\sin^{-1}(x)$ outputs values in $[-\pi/2,\ \pi/2]$, the answer must be in $[-\pi/2,\ \pi/2]$
- **Idea:** Let $\theta = \sin^{-1}(1/2)$. Need to compute θ.
- Take sin of both sides: $\sin\theta = \sin(\sin^{-1}(1/2)) = 1/2$ by the cancellation rule.
- There are many angles θ that work. We want the one in the interval $[-\pi/2,\ \pi/2]$.
- Thus, $\theta = \pi/6$ by the special triangle and SOH CAH TOA.

$$\sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{6}.$$

Arcsine Example

Example

Find $\sin^{-1}(\sin(9\pi/4))$.

Solution:

- By the periodicity of $\sin x$, we have

$$\sin(9\pi/4) = \sin(9\pi/4 - 2\pi) = \sin(\pi/4).$$

- Then, since $\pi/4 \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, by the cancellation rules:

$$\sin^{-1}\left(\sin\left(\frac{9\pi}{4}\right)\right) = \sin^{-1}\left(\sin\left(\frac{\pi}{4}\right)\right) = \frac{\pi}{4}.$$

Arccos Example

Example

Find $\cos^{-1}(0)$, $\cos^{-1}(1)$ and $\cos^{-1}(-1)$.

Solution:

- These simply comes from the graph:

$$\cos^{-1}(0) = \frac{\pi}{2} \qquad \cos^{-1}(1) = 0 \qquad \cos^{-1}(-1) = \pi$$

Example

Find $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$.

Solution:

- **Range**: $\cos^{-1}(x)$ outputs values in $[0,\ \pi]$, thus the answer must be in this interval.
- **Idea:** Let $\theta = \cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$. Need to compute θ.
- Take cos of both sides: $\cos\theta = \cos\left(\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)\right) = \frac{\sqrt{3}}{2}$ by the cancellation rule.
- There are many angles θ that work. We want the one in the interval $[0,\ \pi]$.
- Thus, $\theta = \pi/6$ by the special triangle and SOH CAH TOA.

$$\cos^{-1}\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{6}.$$

Arccos Example

Example

Find $\cos^{-1}(\cos(-5\pi/3))$.

Solution:

- By the periodicity of $\cos x$, we have

$$\cos\left(-\frac{5\pi}{3}\right) = \cos\left(2\pi - \frac{5\pi}{3}\right) = \cos\left(\frac{\pi}{3}\right).$$

- Then, since $\pi/3 \in [0, \pi]$, by the cancellation rule:

$$\cos^{-1}\left(\cos\left(-\frac{5\pi}{3}\right)\right) = \cos^{-1}\left(\cos\left(\frac{\pi}{3}\right)\right) = \frac{\pi}{3}.$$

Arctan Example

Example

Find $\tan^{-1}(\sqrt{3})$.

Solution:

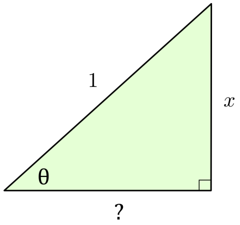
- **Range**: $\tan^{-1}(x)$ outputs values in $(-\pi/2,\ \pi/2)$, thus the answer must be in this interval.
- **Idea:** Let $\theta = \tan^{-1}(\sqrt{3})$. Need to compute θ.
- Take tan of both sides: $\tan\theta = \tan(\tan^{-1}(\sqrt{3})) = \sqrt{3}$ by the cancellation rule.
- There are many angles θ that work. We want the one in the interval $(-\pi/2,\ \pi/2)$.
- Thus, $\theta = \pi/3$ by the special triangle and SOH CAH TOA.

Example

Rewrite the expression $\cos(\sin^{-1} x)$ without trig functions.
(Note that the domain of this function is all $x \in [-1, 1]$.)

Solution:

- **Idea:** Let $\theta = \sin^{-1} x$. Need to compute $\cos\theta$.
- Take sin of both sides: $\sin\theta = \sin(\sin^{-1}(x)) = x$ by the cancellation rule.
- Use SOH CAH TOA and a right triangle (sin is opp/hyp and $\sin\theta = x/1$):



- If z is the remaining side, then by the **Pythagorean Theorem**:

$$z^2 + x^2 = 1 \qquad \to \qquad z^2 = 1 - x^2 \qquad \to \qquad z = \pm\sqrt{1 - x^2}$$

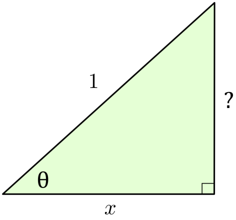
- Since $\theta \in [-\pi/2, \pi/2]$ (since $\theta = \sin^{-1} x$), we must have $\cos\theta \geq 0$.
- Thus, $\cos\theta = \sqrt{1 - x^2}$ by **SOH CAH TOA**, so, $\cos(\sin^{-1} x) = \sqrt{1 - x^2}$

Example

Rewrite the expression $\sin(\cos^{-1} x)$ without trig functions.
(Note that the domain of this function is all $x \in [-1, 1]$.)

Solution:

- **Idea:** Let $\theta = \cos^{-1} x$. Need to compute $\sin\theta$.
- Take cos of both sides: $\cos\theta = \cos(\cos^{-1}(x)) = x$ by the cancellation rule.
- Use SOH CAH TOA and a right triangle (cos is adj/hyp and $\cos\theta = x/1$):



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